

Brain-derived estrogens facilitate male-typical behaviors by potentiating androgen receptor signaling in medaka

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This is an overall **valuable** set of findings on the role of centrally produced estrogens in the control of behaviors in male and female medaka. The significance of the findings rests on the revealed potential mechanism between brain derived estrogens modulating social behaviors in males as well as females. The results are supported by the analysis of multiple transgenic lines although the evidence is **incomplete**, and further validation would be necessary to fully validate the conclusions on the role of brain-derived estrogens. Nonetheless, the findings have led to helpful hypotheses on the hormonal control of behaviors in teleosts that can be tested further.

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Abstract

In rodents, estrogens aromatized from androgens in the brain are essential for the development of male-typical behaviors. In many other vertebrates including humans and teleost fish, however, androgens facilitate these behaviors directly via the androgen receptor without aromatization into estrogens. Here we report that mutagenesis-derived male medaka fish lacking Cyp19a1b (a subtype of aromatase predominantly expressed in the brain) exhibit severely impaired male-typical mating and aggression, despite elevated brain androgen levels. These phenotypes can be rescued by estrogen administration, indicating that brain-derived estrogens are pivotal for male-typical behaviors even in teleosts. Our results further suggest that these estrogens facilitate male-typical behaviors by potentiating androgen action in the brain via the direct stimulation of androgen receptor transcription, thereby revealing a previously unappreciated mode of action of brain-derived estrogens. We additionally show that female fish lacking Cyp19a1b are less receptive to male courtship and conversely court other females, highlighting the significance of brain-derived estrogens in establishing sex-typical behaviors in both sexes.

Introduction

Male and female animals exhibit differences in many innate behaviors, such as mating and aggression (1). In vertebrates, these differences are driven primarily by the influence of sex steroid hormones, including estrogens and androgens. Extensive research in rodents has established that estrogens, traditionally considered “female” hormones, are critical for the development of male-typical behaviors (2–4). Specifically, androgens secreted by the testis, both perinatally and in adulthood, are converted to estrogens in the brain by the enzyme aromatase. These estrogens then act through ESR1, a subtype of the estrogen receptor (ESR), to elicit male-typical behaviors (2–4). This process, originally referred to as the “aromatization hypothesis”, is now widely acknowledged; however, the mechanisms through which brain-derived estrogens affect male-typical behavior, including the identity of their downstream targets, remain largely elusive (4, 5).

More importantly, the aromatization hypothesis seems to apply to only a limited number of species besides rodents (e.g., some birds such as zebra finches) (6). In other species such as humans, other primates, and teleost fish, testicular androgens facilitate male-typical behaviors directly through the androgen receptor (AR) without aromatization (7–9). In primates, the hypothalamic aromatization of androgens to estrogens plays a central role in female gametogenesis (10) but is not essential for male behaviors (7, 8). Notably, in teleosts, 11-ketotestosterone (11KT), which cannot be aromatized to estrogens, is the primary testicular androgen, and exogenous 11KT effectively induces male-typical courtship and aggression even in females (9, 11, 12). Therefore, it is generally assumed that, in teleosts, androgens are largely responsible for male-typical behaviors, while estrogens are dispensable for these behaviors. This is consistent with recent observations in a few teleost species that genetic ablation of AR severely impairs male-typical behaviors (13–16) and with findings in medaka fish (*Oryzias latipes*) that estrogens act through the ESR subtype ESR2b to prevent females from engaging in male-typical courtship (12).

Interestingly, despite these findings, adult teleost brains have extremely high levels of aromatase activity (100–1000 times higher than in rodent brains), resulting in large amounts of brain estrogens even in males (17). In teleost brains, including those of medaka, aromatase is exclusively localized in radial glial cells, in contrast to its neuronal localization in rodent brains (18–20). These observations suggest that brain-derived estrogens have a vital, but as yet undetermined, role in male teleosts. It is worth mentioning that systemic administration of estrogens and an aromatase inhibitor increased and decreased male aggression, respectively, in several teleost species, potentially reflecting the behavioral effects of brain-derived estrogens (21–24). Most teleost species have two distinct genes encoding aromatase, *cyp19a1a* and *cyp19a1b*, due to a whole-genome duplication that occurred early in teleost evolution (17, 25). These genes have undergone subfunctionalization through the partitioning of tissue-specific expression patterns: *cyp19a1a* is expressed predominantly in the gonad, whereas *cyp19a1b* is expressed in the brain (17, 25). In medaka, *cyp19a1b* is also expressed in the gonads, but only at a level tens to hundreds of times lower than in the brain and substantially lower than that of *cyp19a1a* (26, 27).

In the present study, we generated *cyp19a1b*-deficient medaka, in which estrogen synthesis in the brain is selectively impaired while that in the gonads remains intact, in order to investigate the impact of brain-derived estrogens on male behaviors. Remarkably, the fish showed severely impaired male-typical behaviors, thus revealing the marked behavioral effects of these estrogens. Our results further suggest that these effects are mediated by potentiating androgen signaling in behaviorally relevant brain regions.

Results

cyp19a1b-deficient males exhibit severely impaired male-typical mating and aggressive behaviors

We generated a *cyp19a1b*-deficient medaka line from a mutant founder carrying a nonsense mutation in exon 4 of *cyp19a1b* that was identified through screening of the medaka TILLING (targeting-induced local lesions in genomes) library (28) (Fig. S1, A and B). Loss of *cyp19a1b* function in these fish was verified by measuring brain and peripheral levels of sex steroids. As expected, brain estradiol-17 β (E2) in both male and female homozygous mutants (*cyp19a1b*^{-/-}) was significantly reduced to 16% and 50%, respectively, of the levels in their wild-type (*cyp19a1b*^{+/+}) siblings ($P = 0.0037$, males; $P = 0.0092$, females) (Fig. 1, A and B). In males, brain E2 in heterozygotes (*cyp19a1b*^{+/-}) was also reduced to 45% of the level in wild-type siblings ($P = 0.0284$) (Fig. 1A), indicating a dosage effect of *cyp19a1b* mutation. In contrast, peripheral E2 levels were unaltered in both *cyp19a1b*^{-/-} males and females (Fig. S1, C and D), consistent with the expected functioning of Cyp19a1b primarily in the brain. Strikingly, brain levels of testosterone, as opposed to E2, increased 2.2-fold in *cyp19a1b*^{-/-} males relative to wild-type siblings ($P = 0.0006$) (Fig. 1A). Similarly, brain 11KT levels in *cyp19a1b*^{-/-} males and females increased 6.2- and 1.9-fold, respectively, versus wild-type siblings ($P = 0.0007$, males; $P = 0.0316$, females) (Fig. 1, A and B). These results show that *cyp19a1b*-deficient fish have reduced estrogen levels coupled with increased androgen levels in the brain, confirming the loss of *cyp19a1b* function. They also suggest that the majority of estrogens in the male brain and half of those in the female brain are synthesized locally in the brain. In addition, peripheral 11KT levels in *cyp19a1b*^{-/-} males and females increased 3.7- and 1.8-fold, respectively ($P = 0.0789$, males; $P = 0.0118$, females) (Fig. S1, C and D), indicating peripheral influence in addition to central effects.

Next, we investigated the mating behavior of *cyp19a1b*-deficient males (Fig. 1C). The mating behavior of medaka follows a stereotypical sequence. It begins with the male approaching and closely following the female (following). The male then performs a courtship display, rapidly swimming in a circular pattern in front of the female. If the female is receptive, the male grasps her with his fins (wrapping), culminating in the simultaneous release of eggs and sperm (spawning) (16). Because 11KT, the primary driver of male-typical behaviors in teleosts, was increased in *cyp19a1b*-deficient males, we predicted that these fish would engage more actively in mating. Nevertheless, *cyp19a1b*^{-/-} males showed significantly longer latencies to initiate wrappings ($P = 0.0033$ versus *cyp19a1b*^{+/+}, $P = 0.0195$ versus *cyp19a1b*^{+/-}) and to spawn ($P = 0.0051$ versus *cyp19a1b*^{+/+}, $P = 0.0195$ versus *cyp19a1b*^{+/-}) (Fig. 1D). These results suggest that they are less motivated to mate, though an alternative interpretation that their cognition or sexual preference may be altered cannot be dismissed.

However, no significant differences were evident in latencies to initiate followings and courtship displays (Fig. 1D); therefore, the possibility that *cyp19a1b*-deficient males are less sexually attractive and less preferred by females could not be ruled out. To ascertain whether *cyp19a1b*-deficient males are indeed less motivated to mate, we further tested their mating behavior using *esr2b*-deficient females, which are unreceptive to male courtship (12), as stimulus females (Fig. 1E); this test eliminates the influence of female receptivity, facilitating an exclusive evaluation of male motivation to mate with females (16). We found that *cyp19a1b*^{-/-} males showed significantly longer latencies to initiate followings ($P = 0.0129$ versus *cyp19a1b*^{+/+}, $P = 0.0060$ versus *cyp19a1b*^{+/-}), courtship displays ($P = 0.0003$ versus *cyp19a1b*^{+/+}, $P = 0.0006$ versus *cyp19a1b*^{+/-}), and wrapping attempts ($P = 0.0009$ versus *cyp19a1b*^{+/+}, $P = 0.0282$ versus *cyp19a1b*^{+/-}) (Fig. 1F). In addition, they exhibited significantly fewer courtship displays ($P < 0.0001$ versus both *cyp19a1b*^{+/+} and *cyp19a1b*^{+/-}) and wrapping attempts ($P < 0.0001$ versus both *cyp19a1b*^{+/+} and *cyp19a1b*^{+/-}) (Fig. 1G). These results thus confirmed that *cyp19a1b*-deficient males are less motivated to mate. In

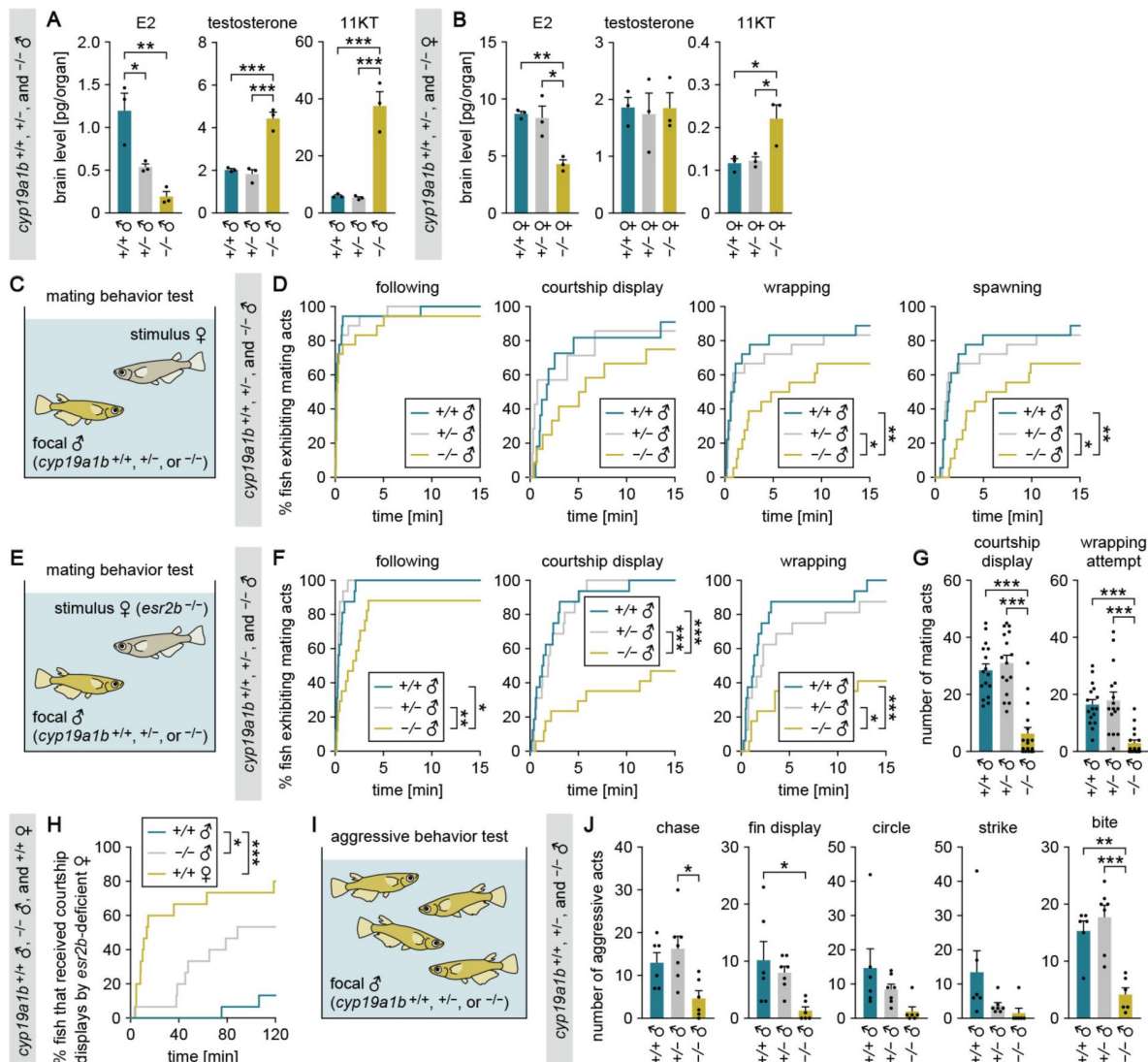


Fig. 1

cyp19a1b-deficient males exhibit severely impaired male-typical mating and aggressive behaviors.

(A and B) Levels of E2, testosterone, and 11KT in the brain of adult *cyp19a1b*^{+/+}, *cyp19a1b*^{+/-}, and *cyp19a1b*^{-/-} males (A) and females (B) (n = 3 per genotype and sex). (C) Set-up for testing the mating behavior of *cyp19a1b*^{+/+}, *cyp19a1b*^{+/-}, and *cyp19a1b*^{-/-} males. (D) Latency of *cyp19a1b*^{+/+}, *cyp19a1b*^{+/-}, and *cyp19a1b*^{-/-} males (n = 18 per genotype) to initiate each mating act toward the stimulus female. (E) Set-up for testing mating behavior using an *esr2b*-deficient female as the stimulus. (F) Latency of *cyp19a1b*^{+/+}, *cyp19a1b*^{+/-}, and *cyp19a1b*^{-/-} males (n = 16, 15, and 17, respectively) to initiate each mating act toward the *esr2b*-deficient female. (G) Number of each mating act performed. (H) Latency of *cyp19a1b*^{+/+} and *cyp19a1b*^{-/-} males and *cyp19a1b*^{+/+} females (n = 15 each) to receive courtship displays from the *esr2b*-deficient female. (I) Set-up for testing aggressive behavior among grouped males. (J) Total number of each aggressive act observed among *cyp19a1b*^{+/+}, *cyp19a1b*^{+/-}, or *cyp19a1b*^{-/-} males in the tank (n = 6, 7, and 5, respectively). Statistical differences were assessed by Bonferroni's or Dunn's post hoc test (A, B, G, and J) and Gehan-Breslow-Wilcoxon test with Bonferroni's correction (D, F, and H). Error bars represent SEM. **P* < 0.05, ***P* < 0.01, ****P* < 0.001.

our previous study, we found that *esr2b*-deficient females court females preferentially over males (12 [Fig. 1H](#)). Consistent with that finding, here we observed that 12 of 15 *cyp19a1b*^{+/+} females received courtship displays from the *esr2b*-deficient female, as compared with only 2 of 15 *cyp19a1b*^{+/+} males ($P = 0.0003$) (**Fig. 1H**). Curiously, we further observed that 8 of 15 *cyp19a1b*^{-/-} males received courtship displays from the *esr2b*-deficient female ($P = 0.0321$ versus *cyp19a1b*^{+/+} males) (**Fig. 1H**). Perhaps *cyp19a1b*^{-/-} males are misidentified as females by *esr2b*-deficient females because they are reluctant to court or they exhibit some female-like behavior.

Next, we examined the intrasexual aggressive behavior of *cyp19a1b*-deficient males (**Fig. 1I**). The aggressive behavior of teleosts including medaka consists of five types of behavioral acts: chases, fin displays, circles, strikes, and bites (29 [Fig. 1J](#)). *cyp19a1b*^{-/-} males exhibited all of these aggressive acts less frequently than *cyp19a1b*^{+/+} and/or *cyp19a1b*^{+/-} males, with significant differences observed for chases ($P = 0.0123$ versus *cyp19a1b*^{+/+}), fin displays ($P = 0.0214$ versus *cyp19a1b*^{+/+}), and bites ($P = 0.0015$ versus *cyp19a1b*^{+/+}, $P = 0.0002$ versus *cyp19a1b*^{+/-}) (**Fig. 1J**). These observations demonstrate that *cyp19a1b*-deficient males are less aggressive toward other males.

In tilapia (*Oreochromis niloticus*), depletion of *cyp19a1b* has been reported to cause male infertility due to efferent duct obstruction (30 [Fig. 1K](#)). If this is also the case in medaka, the observed behavioral defects might be secondary to infertility, possibly due to the perception of impaired sperm release. We therefore investigated the fertilization and hatching success of embryos derived from mating between *cyp19a1b*^{-/-} males and wild-type females. Their fertilization and hatching rates were 88.5% ($n = 183$) and 93.2% ($n = 162$), respectively, indicating that *cyp19a1b*-deficient male medaka have normal fertility.

Brain-derived estrogens facilitate male-typical behaviors probably by stimulating brain AR expression

We considered that the impaired male-typical behaviors of *cyp19a1b*-deficient males might be reasonably attributed to either reduced E2 or increased 11KT in the brain. The latter possibility seemed unlikely because 11KT/AR signaling strongly promotes male-typical behaviors in teleosts (9 [Fig. 2A](#), 13 [Fig. 2B](#)–16 [Fig. 2C](#)); therefore, we tested the former possibility by examining whether E2 treatment would rescue the behavioral phenotypes of *cyp19a1b*-deficient males (**Fig. 2A**). E2 treatment, while having no effect in *cyp19a1b*^{+/+} males (**Fig. 2, B** [Fig. 2C](#) and **Fig. 2D**), significantly shortened the latency to ($P = 0.0005$) and increased the number of ($P = 0.0006$) courtship displays in *cyp19a1b*^{-/-} males (**Fig. 2, D** [Fig. 2E](#) and **Fig. 2F**). These results suggest that reduced E2 in the brain is the primary cause of the mating defects, highlighting a pivotal role of brain-derived estrogens in male mating behavior. However, caution is warranted, as an indirect peripheral effect of bath-immersed E2 on behavior cannot be ruled out, although this is unlikely given the comparable peripheral E2 levels in *cyp19a1b*-deficient and wild-type males. In contrast to mating, E2 treatment was not effective in restoring aggression in *cyp19a1b*^{-/-} males (**Fig. S2A**). It is possible that the treatment protocol used may have failed to replicate the estrogenic milieu necessary to induce aggression in males.

We then considered the mechanism by which brain-derived estrogens facilitate male-typical behaviors. Our previous study showed that exogenous E2 upregulates the expression of a subtype of AR, Ara, in the medaka brain (31; note that Ara was termed Arb in this reference). We therefore hypothesized that brain-derived estrogens may facilitate male-typical behaviors by stimulating Ara expression and thereby potentiating 11KT/Ara signaling in the brain. To test this hypothesis, we first examined *ara* expression in the brain of *cyp19a1b*-deficient males by in situ hybridization analysis. Expression of *ara* was significantly lower in *cyp19a1b*^{-/-} males than in *cyp19a1b*^{+/+} males in several preoptic and hypothalamic nuclei that are activated upon mating and/or attack in males (15 [Fig. 2, F](#) [Fig. 2G](#) and **Fig. 2H**; see Table S1 for abbreviations of brain nuclei). We then performed a similar analysis for the other AR subtype, Arb. Expression of *arb* was significantly lower in *cyp19a1b*^{-/-} than in

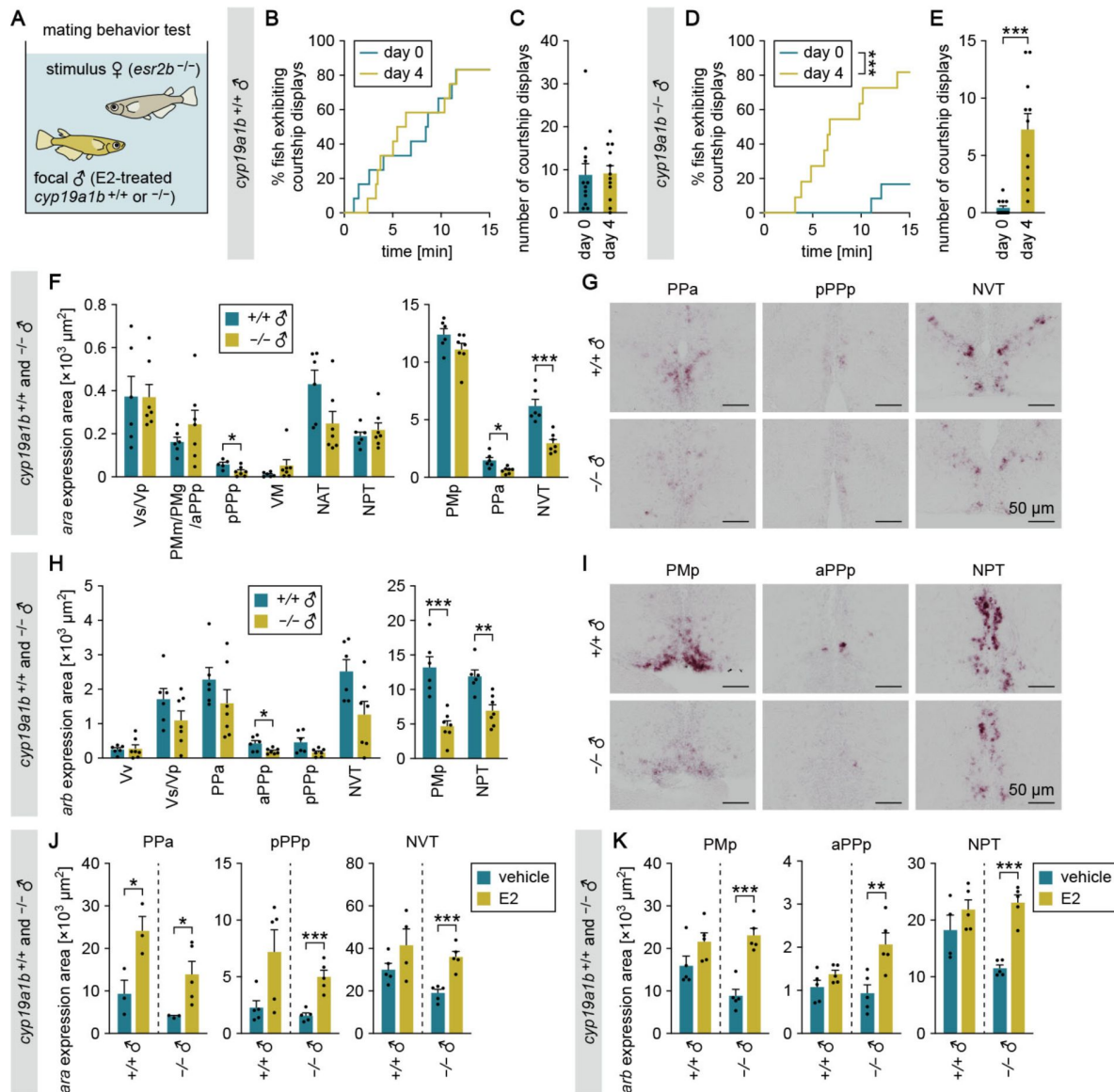


Fig. 2

Brain-derived estrogens facilitate male-typical behaviors probably by stimulating brain AR expression.

(A) Set-up for testing the mating behavior of E2-treated *cyp19a1b*^{+/+} and *cyp19a1b*^{-/-} males. (B) Latency of *cyp19a1b*^{+/+} males (n = 12) to initiate courtship displays toward the stimulus female before (day 0) and after (day 4) E2 treatment. (C) Number of courtship displays performed by *cyp19a1b*^{+/+} males. (D) Latency of *cyp19a1b*^{-/-} males (n = 12) to initiate courtship displays before (day 0) and after (day 4) E2 treatment. (E) Number of courtship displays performed by *cyp19a1b*^{-/-} males. (F) Total area of *ara* expression signal in each brain nucleus of *cyp19a1b*^{+/+} (n = 6 except for pPPp, where n = 5) and *cyp19a1b*^{-/-} (n = 7) males. The data are displayed in two graphs for visual clarity. (G) Representative images of *ara* expression in the PPa, pPPp, and NVT. (H) Total area of *arb* expression signal in each brain nucleus of *cyp19a1b*^{+/+} (n = 6) and *cyp19a1b*^{-/-} (n = 7 except for NVT, where n = 6) males. The data are displayed in two graphs for visual clarity. (I) Representative images of *arb* expression in the PMp, aPPp, and NPT. (J) Total area of *ara* expression signal in the PPa, pPPp, and NVT of *cyp19a1b*^{+/+} and *cyp19a1b*^{-/-} males treated with vehicle alone or E2 (n = 5 per group except for NVT of E2-treated *cyp19a1b*^{+/+} males, where n = 4; and PPa of vehicle-treated *cyp19a1b*^{+/+}, E2-treated *cyp19a1b*^{+/+}, and vehicle-treated *cyp19a1b*^{-/-} males, where n = 3). (K) Total area of *arb* expression signal in the PMp, aPPp, and NPT of *cyp19a1b*^{+/+} and *cyp19a1b*^{-/-} males treated with vehicle alone or E2 (n = 5 per group except for NPT of vehicle-treated *cyp19a1b*^{+/+} males, where n = 4). Scale bars represent 50 μm. For abbreviations of brain nuclei, see Table S1. Statistical differences were assessed by Gehan-Breslow-Wilcoxon test (B and D) and unpaired *t* test, with Welch's correction where appropriate (C, E, F, H, J, and K). Error bars represent SEM. **P* < 0.05, ***P* < 0.01, ****P* < 0.001.

cyp19a1b^{+/+} males in other preoptic and hypothalamic nuclei activated upon mating and/or attack (16 [↗](#)), including the PMp, aPPp, and NPT ($P = 0.0009$, 0.0413 , and 0.0021 , respectively) (Fig. 2, H [↗](#) and I [↗](#)).

Next, to determine whether these reductions in *ara* and *arb* expression in *cyp19a1b*^{-/-} males were the result of reduced brain E2, we tested whether E2 treatment could restore the expression of the two genes. In situ hybridization revealed significantly increased *ara* and *arb* expression in most brain nuclei of E2-treated *cyp19a1b*^{-/-} males as compared with vehicle-treated *cyp19a1b*^{-/-} males (similar results were obtained in *cyp19a1b*^{+/+} males) (Fig. 2, J [↗](#) and K [↗](#); Fig. S2, B and C; Fig. S3), indicating that the decreased *ara* and *arb* expression in *cyp19a1b*^{-/-} males is attributable to reduced E2 levels. Taken together, these results suggest that brain-derived estrogens elicit male-typical behaviors by stimulating *ara* and *arb* expression in behaviorally relevant brain regions.

***cyp19a1b* deficiency impairs behaviorally relevant signaling pathways downstream of ARs**

We considered that, if our hypothesis that brain-derived estrogens facilitate male-typical behaviors by potentiating androgen/AR signaling is correct, then *cyp19a1b*-deficient males should have impaired activation of behaviorally relevant genes that act downstream of ARs. Two neuropeptide genes, *vt* (encoding vasotocin) and *gal* (encoding galanin), have been implicated in male-typical mating and aggressive behaviors in various vertebrates, including medaka and other teleosts (11 [↗](#), 16 [↗](#), 29 [↗](#), 32 [↗](#), 33 [↗](#)). In medaka, subsets of neurons in the pNVT and pPMp express *vt* and *gal*, respectively, in an androgen-dependent and hence male-biased manner, and these neurons have been implicated in male-typical behaviors (11 [↗](#), 29 [↗](#), 34 [↗](#)). We therefore studied the expression of *vt* and *gal* in the brains of *cyp19a1b*-deficient males.

In situ hybridization revealed that, as expected, expression of *vt* in the pNVT of *cyp19a1b*^{-/-} males was significantly reduced to 18% as compared with *cyp19a1b*^{+/+} males ($P = 0.0040$), a level comparable to that observed in females (Fig. 3, A [↗](#) and B [↗](#)). In contrast, there were no significant differences between genotypes in other brain nuclei (Fig. 3A [↗](#)). Similarly, expression of *gal* in the pPMp of *cyp19a1b*^{-/-} males was reduced to 43% as compared with *cyp19a1b*^{+/+} males ($P = 0.0019$), while no significant differences were observed in other brain nuclei (Fig. 3, C [↗](#) and D [↗](#)). These results demonstrate that *cyp19a1b* deficiency severely impairs the AR-dependent activation of behaviorally relevant *vt* and *gal* expression, suggesting that brain-derived estrogens play a substantial role in activating AR signaling.

Estrogens directly stimulate the transcription of ARs through ESRs

The above results led us to further explore how brain-derived estrogens stimulate the expression of *ara* and *arb*. In silico analysis of the medaka *ara* and *arb* loci identified two canonical bipartite estrogen-responsive element (ERE)-like sequences in introns 1 and 2 of *ara* (located at positions +1699 and +2050, respectively, relative to the transcription initiation site) and three in the 5'-flanking region of *arb* (at positions -1272, -2180, and -3327) (Fig. 4, A [↗](#) and B [↗](#)). Thus, brain estrogens might be able to directly activate the transcription of *ara* and *arb*. To evaluate this likelihood, we conducted in vitro transcriptional activity assays in which luciferase expression was driven by genomic fragments from the *ara* and *arb* loci containing the identified ERE-like sequences.

An assay using the *ara* genomic fragment revealed that E2 dose-dependently increased luciferase activity in the presence of any ESR subtype (Esr1, Esr2a, or Esr2b) (Fig. 4C [↗](#)), suggesting that E2 has the ability to directly activate *ara* transcription through ESRs. The introduction of a point mutation into the ERE-like sequence at position +2050 abolished the E2-induced luciferase activity in the presence of any ESR subtype, while mutation at +1699 had no such effect (Fig. 4D [↗](#)). These results suggest that the ERE at position +2050 is responsible for E2-induced activation of *ara*.

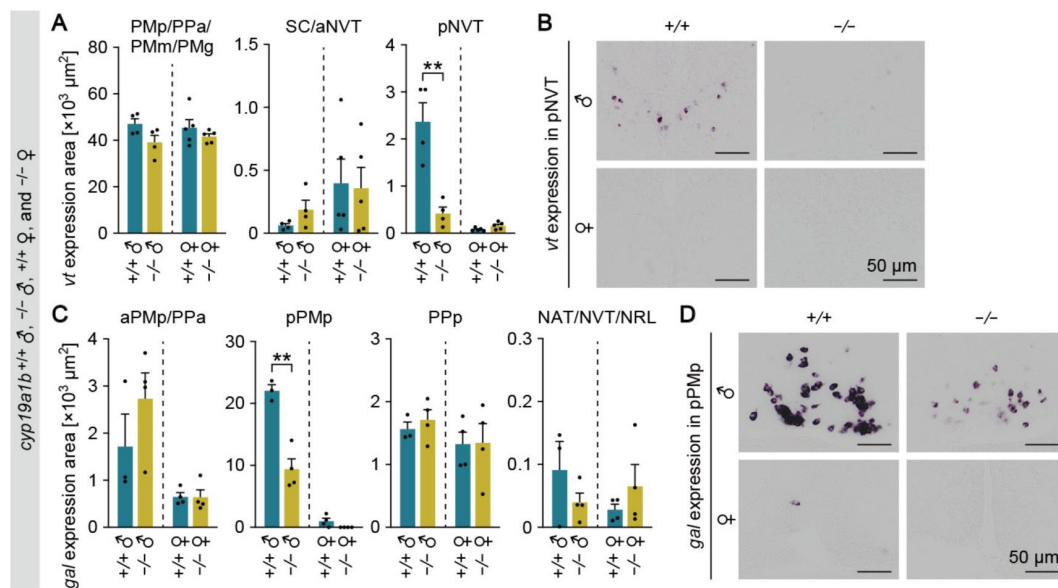


Fig. 3

***cyp19a1b* deficiency impairs behaviorally relevant signaling pathways downstream of ARs.**

(A) Total area of *vt* expression signal in the PMP/PPa/PMm/PMg, SC/aNVT, and pNVT of *cyp19a1b*^{+/+} and *cyp19a1b*^{-/-} males and females (n = 4 and 5 per genotype for males and females, respectively). (B) Representative images of *vt* expression in the pNVT. (C) Total area of *gal* expression signal in the aPMp/PPa, pPMp, PPp, and NAT/NVT/NRL of *cyp19a1b*^{+/+} and *cyp19a1b*^{-/-} males and females (n = 4 except for *cyp19a1b*^{+/+} males, where n = 3). (D) Representative images of *gal* expression in the pPMp. Scale bars represent 50 μm . For abbreviations of brain nuclei, see Table S1. Statistical differences were assessed by unpaired *t* test, with Welch's correction where appropriate (A and C). Error bars represent SEM. ***P* < 0.01.

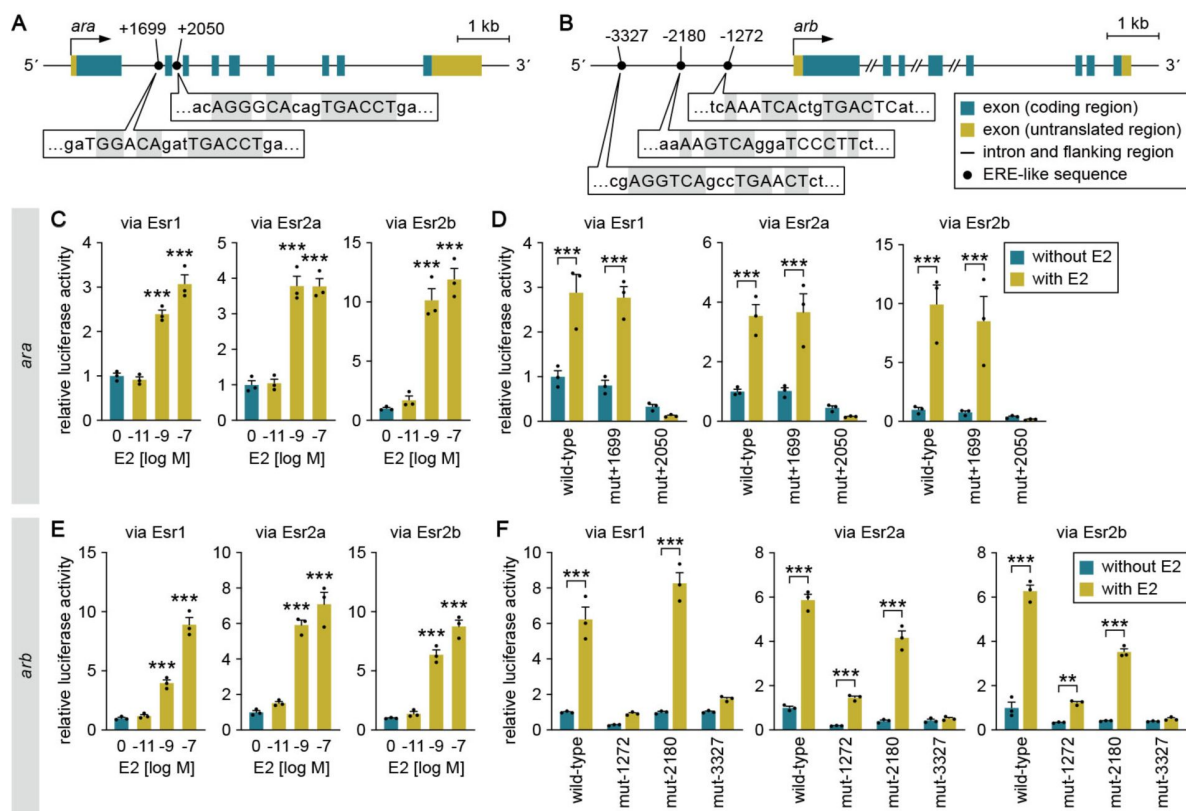


Fig. 4

Estrogens directly stimulate the transcription of ARs through ESRs.

(**A** and **B**) Schematic of *ara* (**A**) and *arb* (**B**) loci showing the location of the canonical bipartite ERE-like sequences. Bent arrows mark the transcription initiation sites. Nucleotides of the ERE-like sequences are denoted by capital letters, and those identical to the consensus ERE (AGGTCAnnnTGACCT) are gray-shaded. (**C**) Ability of E2 to directly activate *ara* transcription. Cultured cells were transfected with a luciferase reporter construct containing a genomic fragment upstream of exon 3 of *ara*, together with an Esr1, Esr2a, or Esr2b expression construct. The cells were stimulated with different concentrations of E2 and luciferase activity was measured. (**D**) Effect of mutations in the ERE-like sequences on the E2-induced activation of *ara* transcription. Cultured cells were transfected with a wild-type luciferase construct or a construct carrying a mutation in the ERE-like sequence at position +1699 (mut+1699) or +2050 (mut+2050), together with an Esr1, Esr2a, or Esr2b expression construct. The cells were stimulated with or without E2, and luciferase activity was measured. (**E**) Ability of E2 to directly activate *arb* transcription. The assay described in **C** was performed with a luciferase construct containing a genomic fragment upstream of the first methionine codon of *arb*. (**F**) Effect of mutations in the ERE-like sequences on the E2-induced activation of *arb* transcription. The assay described in **D** was performed with luciferase constructs each carrying a mutation in the ERE-like sequence at position -1272 (mut-1272), -2180 (mut-2180), or -3327 (mut-3327). Values are expressed as a fold change relative to a control without E2 stimulation (**C** and **E**) or a control using the wild-type construct without E2 stimulation (**D** and **F**). Statistical differences were assessed by Dunnett's post hoc test (**C** and **E**) and unpaired *t* test with Bonferroni-Dunn correction (**D** and **F**). Error bars represent SEM. ***P* < 0.01; ****P* < 0.001.

transcription. Because there is evidence that a single ERE half-site is sufficient to confer E2 responsiveness on several genes (35), we performed the assay using *ara* genomic fragments in which a mutation was introduced in either half-site of the ERE at +2050. E2 induction of luciferase activity was abrogated in both cases (Fig. S4A), suggesting that both ERE half-sites are required to confer estrogen responsiveness on *ara*.

In the assay using the *arb* genomic fragment, E2 also dose-dependently increased luciferase activity in the presence of any ESR subtype (Fig. 4E). The E2-induced increase was completely abolished by point mutation of the ERE-like sequence at position -3327 in the presence of any ESR subtype (Fig. 4F). A similar effect was observed for mutation of the ERE-like sequence at position -1272, but only in the presence of Esr1 and not in the presence of Esr2a or Esr2b (Fig. 4F). These results indicate that E2 can directly stimulate *arb* transcription, primarily through the ERE at -3327. Mutations in either half-site of this ERE eliminated induction by E2 in the presence of Esr2a or Esr2b, but not Esr1 (Fig. S4B), suggesting that both ERE half-sites are required to confer estrogen responsiveness on *arb*. Collectively, these experiments suggest that the transcription of *ara* and *arb* can be directly stimulated by estrogens (including brain-derived estrogens) via the binding of ESRs to canonical bipartite EREs.

Brain-derived estrogens stimulate *ara* and *arb* expression in behaviorally relevant brain regions primarily through Esr2a and Esr1, respectively

Next, we investigated whether brain-derived estrogens can induce *ara* and *arb* expression through ESRs in vivo and, if so, which ESR subtype (Esr1, Esr2a, or Esr2b) mediates this induction. To this end, we examined *ara* and *arb* expression in the brains of males deficient for each ESR subtype by in situ hybridization. We used previously described *esr1*- and *esr2b*-deficient medaka (12, 36) and generated *esr2a*-deficient medaka using the CRISPR (clustered regularly interspaced short palindromic repeats) / Cas9 (CRISPR-associated protein 9) system. Two independent *esr2a*-deficient lines ($\Delta 8$ and $\Delta 4$) were established (Fig. S5) and used for subsequent behavioral experiments to ensure the reproducibility of the observed phenotypes.

In *esr1*^{+/+} and *esr1*^{-/-} males, *ara* expression was not significantly different between genotypes in any brain nucleus (Fig. S6A), but *arb* expression was significantly lower in *esr1*^{-/-} than in *esr1*^{+/+} males in the PMp and aPPp ($P = 0.0111$ and 0.0376 , respectively) (Fig. 5, A and B). This, together with our observation that *arb* expression in these brain nuclei was also significantly lower in *cyp19a1b*^{-/-} males (Fig. 2H), suggests that brain-derived estrogens stimulate *arb* expression in these nuclei primarily through Esr1. In support of this notion, double-label in situ hybridization in the wild-type male brain detected neurons coexpressing *arb* and *esr1* in both the PMp and aPPp (Fig. S6B). In *esr2a*^{+/+} and *esr2a*^{-/-} male brains (from the $\Delta 8$ line), *ara* expression was significantly lower in *esr2a*^{-/-} than in *esr2a*^{+/+} males in the PPa, pPPp, and NVT ($P = 0.0359$, 0.0430 , and 0.0178 , respectively) (Fig. 5, C and D), while no significant difference was observed in *arb* expression (Fig. S6C). Considering that *ara* expression in these nuclei was also lower in *cyp19a1b*^{-/-} males (Fig. 2F), brain-derived estrogens presumably stimulate *ara* expression in these nuclei mainly through Esr2a. This idea was further supported by double-label in situ hybridization, which detected neurons coexpressing *ara* and *esr2a* in the PPa, pPPp, and NVT of wild-type males (Fig. S6D). Examination of *esr2b*^{+/+} and *esr2b*^{-/-} male brains showed significantly lower expression of *ara* in the NAT and higher expression of *arb* in the PPa of *esr2b*^{-/-} males (Fig. S6, E to H); however, no such changes in expression were observed in *cyp19a1b*^{-/-} males (Fig. 2, F and H). It therefore seems unlikely that Esr2b mediates the effect of brain estrogens on AR expression. Collectively, these results suggest that brain-derived estrogens stimulate *ara* and *arb* expression in behaviorally relevant brain regions primarily through Esr2a and Esr1, respectively.

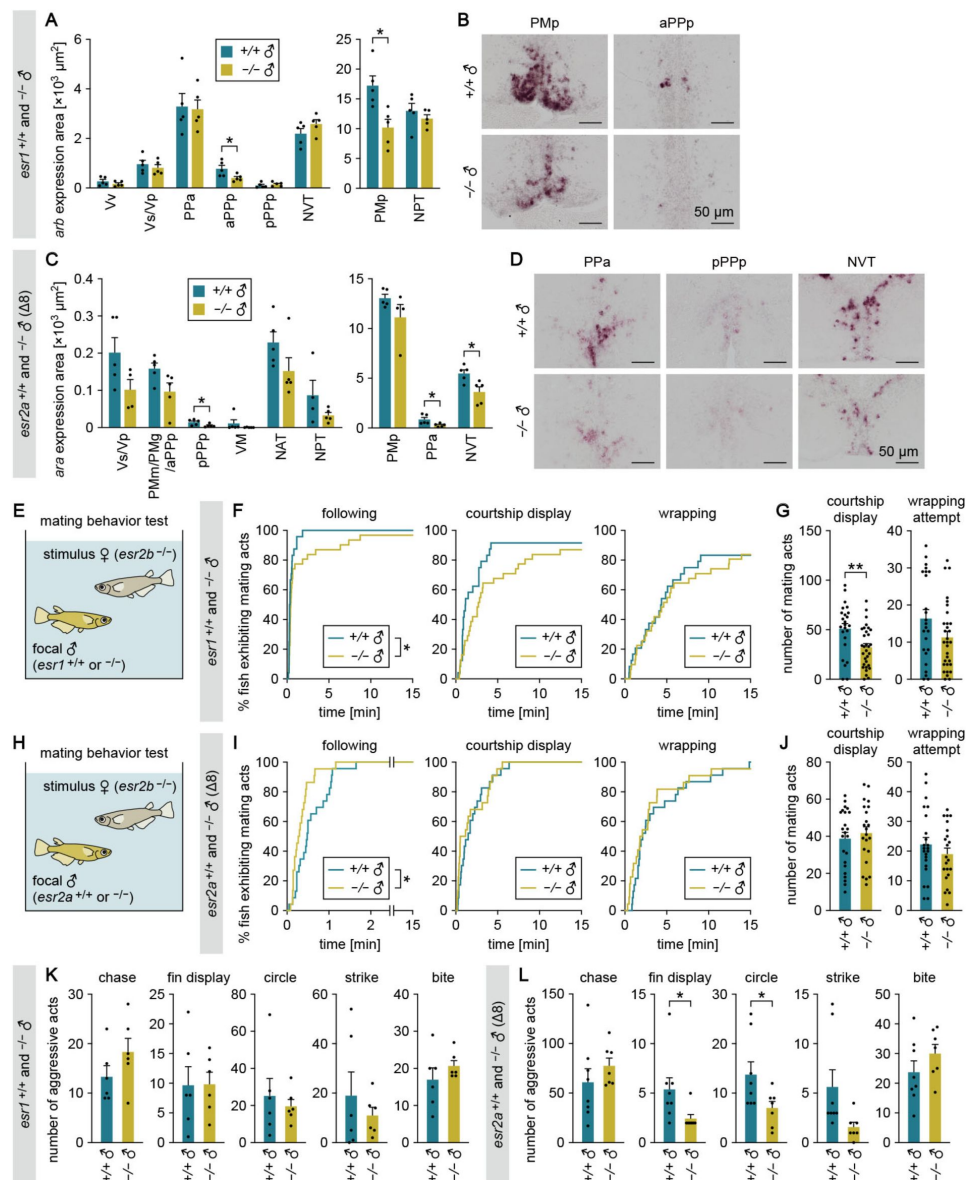


Fig. 5

Brain-derived estrogens stimulate *ara* and *arb* expression in behaviorally relevant brain regions primarily through *Esr2a* and *Esr1*, respectively.

(A) Total area of *arb* expression signal in each brain nucleus of *esr1*^{+/+} and *esr1*^{-/-} males (n = 5 per genotype). The data are displayed in two graphs for visual clarity. (B) Representative images of *arb* expression in the PMp and aPPp. (C) Total area of *ara* expression signal in each brain nucleus of *esr2a*^{+/+} and *esr2a*^{-/-} males ($\Delta 8$ line; n = 5 per genotype except for NPT of *esr2a*^{+/+} males and Vs/Vp and PMp of *esr2a*^{-/-} males, where n = 4). The data are displayed in two graphs for visual clarity. (D) Representative images of *ara* expression in the PPa, pPPp, and NVT. (E) Set-up for testing the mating behavior of *esr1*^{+/+} and *esr1*^{-/-} males using an *esr2b*-deficient female as the stimulus. (F) Latency of *esr1*^{+/+} and *esr1*^{-/-} males (n = 24 and 31, respectively) to initiate each mating act toward the stimulus female. (G) Number of each mating act performed. (H) Set-up for testing the mating behavior of *esr2a*^{+/+} and *esr2a*^{-/-} males using an *esr2b*-deficient female as the stimulus. (I) Latency of *esr2a*^{+/+} and *esr2a*^{-/-} males ($\Delta 8$ line; n = 23 and 22, respectively) to initiate each mating act toward the stimulus female. (J) Number of each mating act performed. (K and L) Total number of each aggressive act observed among *esr1*^{+/+} or *esr1*^{-/-} males (n = 6 per genotype) (K) and among *esr2a*^{+/+} or *esr2a*^{-/-} males ($\Delta 8$ line; n = 8 and 7, respectively) (L) in the tank. Scale bars represent 50 μ m. For abbreviations of brain nuclei, see Table S1. Statistical differences were assessed by unpaired *t* test, with Welch's correction where appropriate (A, C, G, J, K, and L) and Gehan-Breslow-Wilcoxon test (F and I). Error bars represent SEM. **P* < 0.05, ***P* < 0.01.

***esr1*- and *esr2a*-deficient males are, respectively, less motivated to mate and less aggressive**

If *Esr1* and *Esr2a* truly mediate the behavioral effects of brain-derived estrogens through the activation of AR expression, then *esr1*- and *esr2a*-deficient males should exhibit impaired male-typical behaviors. More specifically, given that *Esr1* and *Esr2a* function to activate the expression of *arb* and *ara*, which are responsible for male mating and aggression, respectively (16), *esr1*- and *esr2a*-deficient males should be less motivated to mate and less aggressive, respectively.

To test these ideas, we first evaluated the mating behavior of *esr1*-deficient males using a wild-type female as the stimulus. As expected, *esr1*^{-/-} males showed a significantly longer latency to initiate followings than *esr1*^{+/+} males ($P = 0.0055$) (Fig. S7), suggesting that they are less motivated to mate. This was further confirmed in tests using an *esr2b*-deficient female as the stimulus, where *esr1*^{-/-} males showed a longer latency to followings and fewer courtship displays than *esr1*^{+/+} males ($P = 0.0426$ and 0.0039 , respectively) (Fig. 5, E to G). In contrast, no deficits were observed in the mating behavior of *esr2a*^{-/-} males toward wild-type females (Fig. S8, A and B). Although *esr2a*^{-/-} males of the $\Delta 8$ line showed a shorter latency to followings than their wild-type siblings in tests using a stimulus *esr2b*-deficient female ($P = 0.0146$) (Fig. 5, H to J), this was not reproduced in the $\Delta 4$ line (Fig. S8, C and D). We subsequently assessed aggressive behavior and found no defects in *esr1*^{-/-} males (Fig. 5K). Conversely, *esr2a*^{-/-} males from both the $\Delta 8$ and $\Delta 4$ lines exhibited significantly fewer fin displays than their wild-type siblings ($P = 0.0461$ and 0.0293 , respectively). Circles followed a similar pattern, with a significant reduction in the $\Delta 8$ line ($P = 0.0446$) and a comparable but non-significant decrease in the $\Delta 4$ line ($P = 0.1512$) (Fig. 5L; Fig. S8E), showing less aggression. Taken together with the previous finding that *esr2b*-deficient males show no deficits in either mating or aggressive behavior (12), these results suggest that brain-derived estrogens can promote mating with females by stimulating *arb* expression through *Esr1* and can increase aggression toward other males by stimulating *ara* expression through *Esr2a*. Nonetheless, behavioral deficits in *esr1*- and *esr2a*-deficient males were relatively mild as compared with *cyp19a1b*-deficient males (Fig. 1) and *ara*- and *arb*-deficient males (16), suggesting that a compensatory mechanism may exist between ESR subtypes.

***cyp19a1b*-deficient females are less receptive to males and instead court other females**

Finally, we tested the behavioral impact of *cyp19a1b* deficiency in females. Tests of the mating behavior of *cyp19a1b*^{-/-} females revealed that they had significantly longer latencies to receive the first wrapping from the stimulus male ($P = 0.0018$ versus *cyp19a1b*^{+/+}, $P = 0.0117$ versus *cyp19a1b*^{+/-}) and to spawn ($P = 0.0015$ versus *cyp19a1b*^{+/+}) (Fig. 6, A and B). These results are consistent with a recent study in zebrafish (*Danio rerio*) showing that *cyp19a1b*-deficient females are less receptive to male courtship (37). We further found that the number of courtship displays and wrapping attempts from the male was not increased in *cyp19a1b*^{-/-} females (Fig. 6C), despite their reduced receptivity. This suggests that they are not only less receptive to male courtship but may also actively avoid males, depriving them of mating opportunities. We tested this assumption by measuring the distance between fish, which confirmed that *cyp19a1b*^{-/-} females spent significantly less time near males at distances of 10–20 mm ($P = 0.0438$) and 20–30 mm ($P = 0.0002$) as compared with *cyp19a1b*^{+/+} females (Fig. 6D). We therefore explored the possibility that *cyp19a1b*^{-/-} females may develop a stronger sexual preference for females than for males. In tests of the mating behavior between two females, 4 of 10 *cyp19a1b*^{-/-} females exhibited courtship displays toward the stimulus female, while 0 of 9 *cyp19a1b*^{+/+} females did ($P = 0.0396$) (Fig. 6, E and F). These results demonstrate that *cyp19a1b*-deficient females, having a slightly female-oriented sexual preference, are less receptive to male courtship and, conversely, court other females.

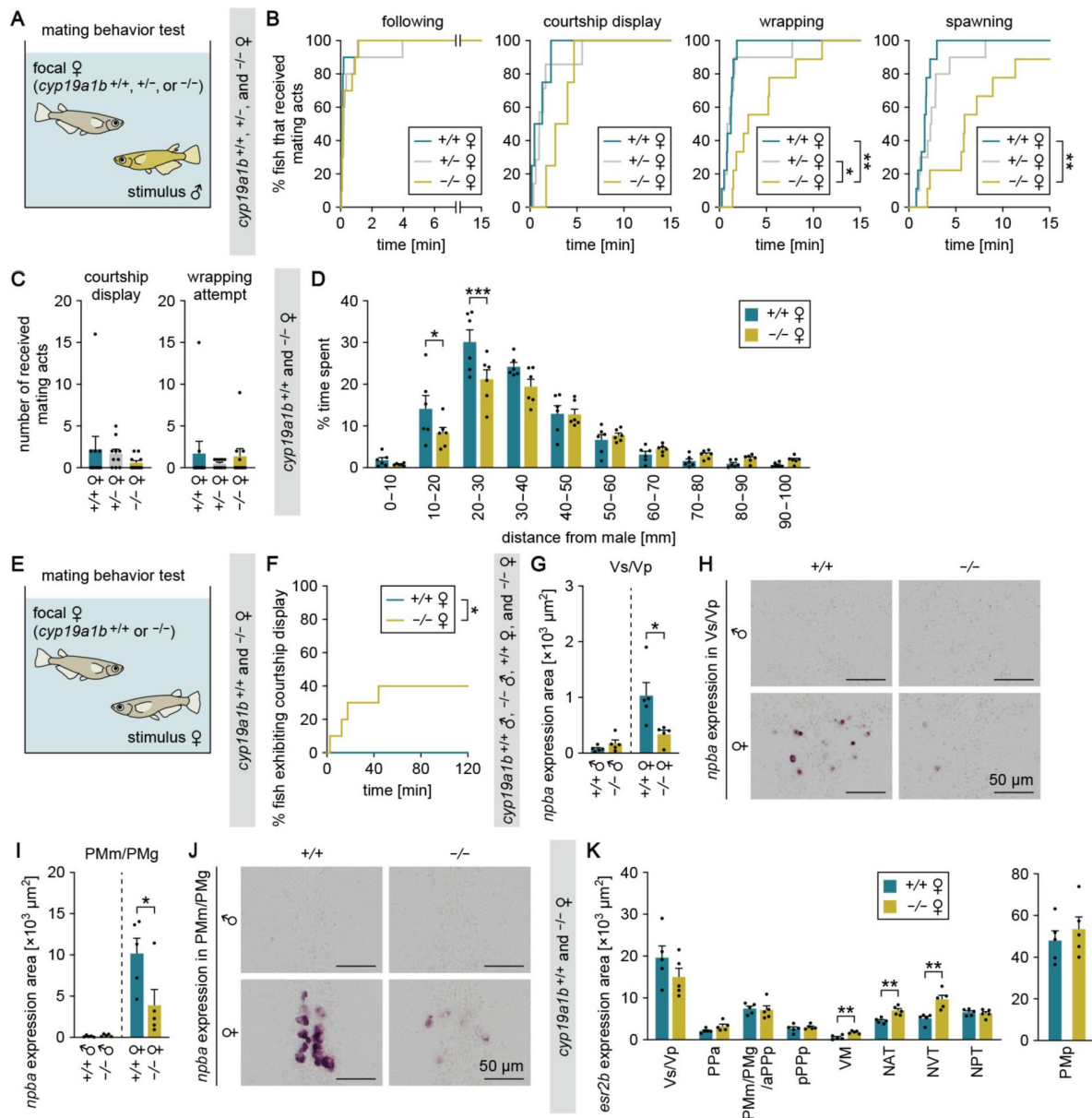


Fig. 6

cyp19a1b-deficient females are less receptive to males and instead court other females.

(A) Set-up for testing the mating behavior of *cyp19a1b*^{+/+}, *cyp19a1b*^{+/-}, and *cyp19a1b*^{-/-} females. (B) Latency of *cyp19a1b*^{+/+}, *cyp19a1b*^{+/-}, and *cyp19a1b*^{-/-} females (n = 10 per genotype) to receive each mating act from the stimulus male. (C) Number of each mating act received. (D) Percentage of time *cyp19a1b*^{+/+} and *cyp19a1b*^{-/-} females (n = 6 per genotype) spent at different distances from the stimulus male. (E) Set-up for testing the mating behavior of *cyp19a1b*^{+/+} and *cyp19a1b*^{-/-} females toward a stimulus female. (F) Latency of *cyp19a1b*^{+/+} (n = 9) or *cyp19a1b*^{-/-} (n = 10) females to initiate courtship displays toward the stimulus female. (G and H) Total area (G) and representative images (H) of *npba* expression signal in the Vs/Vp of *cyp19a1b*^{+/+} and *cyp19a1b*^{-/-} males and females (n = 5 per genotype and sex). (I and J) Total area (I) and representative images (J) of *npba* expression signal in the PMm/PMg of *cyp19a1b*^{+/+} and *cyp19a1b*^{-/-} males and females (n = 5 per genotype and sex). (K) Total area of *esr2b* expression signal in each brain nucleus of *cyp19a1b*^{+/+} or *cyp19a1b*^{-/-} females (n = 5 per genotype). The data are displayed in two graphs for visual clarity. Scale bars represent 50 μm . For abbreviations of brain nuclei, see Table S1. Statistical differences were assessed by Gehan-Breslow-Wilcoxon test, with Bonferroni's correction where appropriate (B and F), Bonferroni's post hoc test (C), unpaired *t* test with Bonferroni-Dunn correction (D), and unpaired *t* test, with Welch's correction where appropriate (G, I, and K). Error bars represent SEM. **P* < 0.05, ***P* < 0.01, ****P* < 0.001.

Notably, these behavioral phenotypes of *cyp19a1b*-deficient females, albeit milder, closely resemble those of *esr2b*-deficient females (12). This led us to hypothesize that *cyp19a1b* deficiency prevents female-typical, and induces male-typical, behavioral phenotypes by decreasing estrogen/Esr2b signaling. If correct, *cyp19a1b*-deficient females should be defective in the activation of behaviorally relevant genes that act downstream of Esr2b. In medaka, subsets of neurons in the Vs/Vp and PMm/PMg express *npba* (encoding neuropeptide B) in an estrogen/Esr2b signaling-dependent and hence female-specific manner, and these neurons have been implicated in female sexual receptivity (12, 38, 39). We therefore examined the expression of *npba* in the brains of *cyp19a1b*-deficient females by in situ hybridization. As expected, *npba* expression in the Vs/Vp and PMm/PMg of *cyp19a1b*^{-/-} females was significantly reduced to 33% and 38%, respectively, of that of *cyp19a1b*^{+/+} females (Vs/Vp, *P* = 0.0386; PMm/PMg, *P* = 0.0461) (Fig. 6, G to J). Expression of *esr2b* in these brain nuclei is also estrogen-dependent and female-specific (12, 31); therefore, we tested whether it is also attributable to brain-derived estrogens. In situ hybridization showed no significant differences in *esr2b* expression between *cyp19a1b*^{+/+} and *cyp19a1b*^{-/-} females in the Vs/Vp and PMm/PMg, although significant differences were seen in the VM, NAT, and NVT (Fig. 6K), suggesting that *esr2b* expression in the brain relies on ovarian estrogens and not on brain-derived estrogens. Taken altogether, our findings indicate that brain-derived estrogens contribute substantially to the induction of female-typical, and the prevention of male-typical, behaviors in females by activating estrogen/Esr2b signaling.

Discussion

In this study, we found that male medaka deficient for *cyp19a1b* exhibit severely impaired male-typical mating and aggression. This observation was noteworthy because the fish had markedly elevated brain levels of 11KT, the primary driver of male-typical behaviors in teleosts (9, 11). Deficits in mating were rescued by estrogen administration, indicating that reduced brain estrogen levels are the primary cause of the observed mating impairment; in other words, brain-derived estrogens are pivotal at least for male-typical mating behaviors in teleosts. This was also notable because, unlike in rodents, where the necessity for brain-derived estrogens has been codified as the “aromatization hypothesis”, these estrogens have been regarded as dispensable for male behaviors in many vertebrates, including teleosts (7–9). In addition, the rescue of behavioral phenotypes by estrogen administration in adults suggests that in teleosts, unlike in rodents, brain-derived estrogens early in life is not essential for the execution of male-typical behaviors. While brain-derived estrogens are necessary for male behaviors in both rodents and teleosts, the life stages at which they exert their behavioral effects probably differ between these species. Brain aromatase activity in teleosts increases with age and, at adulthood, reaches 100–1000 times that in rodents (9, 17). In contrast, brain aromatase activity in rodents reaches its peak during the perinatal period and thereafter declines with age (40), although it remains important for male behavior in adulthood. This variation among species may represent the activation of brain estrogen synthesis at life stages critical for sexual differentiation of behavior that are unique to each species.

Brain-derived estrogens serve several functions during the process of sexual differentiation of the mouse brain, including the synthesis of prostaglandin PGE2 and the activation of synaptic and neurodevelopmental genes; however, the specific mechanism whereby these estrogens affect male behaviors remains obscure (4, 5, 41). Our findings in medaka indicate that brain-derived estrogens facilitate male-typical behaviors by potentiating androgen/AR signaling in behaviorally relevant brain regions via the direct stimulation of AR transcription. More specifically, they indicate that brain-derived estrogens (1) promote mating by stimulating the transcription of *arb* in some preoptic and hypothalamus nuclei via Esr1, and (2) increase aggression by stimulating the transcription of *ara* in other preoptic and hypothalamus nuclei through Esr2a. This model of the mechanism underlying the action of brain-derived estrogens would explain the apparent contradiction that, in teleosts, androgens per se elicit male behaviors without aromatization, while

estrogen synthesis in the brain are also critical for these behaviors. Our data also indicate that the two AR genes, *ara* and *arb*, are direct downstream targets of brain-derived estrogens that mediate male behaviors, only a few of which have been identified thus far. In mice, perinatal brain estrogens increase *Ar* expression in the bed nucleus of the stria terminalis and preoptic area, two brain regions that have been implicated in male behaviors (42). Recent evidence suggests that ESR1 binds to the regulatory genomic region of *Ar* in these brain regions in mice (41). Given these facts, the idea that brain-derived estrogens enhance androgen/AR signaling by directly stimulating AR transcription may apply to a wide range of species, including rodents.

Consistent with our results, studies in several teleost species have shown that treatment of males with an aromatase inhibitor reduces their male-typical behaviors, while estrogens exert the opposite effect (21–24). Conversely, it has been shown in various teleosts, including medaka, that treatment with exogenous estrogens attenuates male behaviors (43–47). A possible explanation for this discrepancy is that estrogens may either stimulate or suppress male-typical behaviors, depending on their concentration. All studies showing the suppressive effects of exogenous estrogens were conducted at doses higher than those used in the present study or at doses mimicking the levels typical of adult females (48). In addition, our previous study in male medaka showed that high doses of exogenous estrogens induce the expression of *esr2b*, which prevents male-typical mating behavior, in behaviorally relevant brain regions (12). Thus, the development of male behaviors may require moderate brain estrogen levels that are sufficient to induce the expression of *ara* and *arb*, but not *esr2b*, in the underlying neural circuitry. This may account for the lack of aggression recovery in E2-treated *cyp19a1b*-deficient males in this study. Another possibility that is not mutually exclusive is that endogenous levels of brain estrogens are sufficient to motivate males to engage in male-typical behaviors, and therefore exogenous estrogens have no further effect. This possibility is at least likely for mating behavior, as estrogen treatment facilitated mating behavior in *cyp19a1b*-deficient males but not in their wild-type siblings. Further studies using *cyp19a1b* mutants from different teleost species are needed to explore these possibilities and to determine whether the findings in medaka hold for other teleosts.

We also found that, as compared with males, females have higher brain levels of estrogens, with half being synthesized locally and the other half supplied by the ovaries. In many brain nuclei of medaka, expression of *ara* and *arb* is comparable between the sexes or at higher levels in females (31), suggesting that brain-derived estrogens stimulate AR expression in females as well as in males. This raises the issue of why females do not display male-typical behaviors despite such high brain levels of estrogens and active estrogen synthesis in the brain. Possibly, as mentioned above, high estrogen levels in the female brain may suppress male-typical behaviors by promoting the expression of *esr2b*. Another possibility is that females, despite having abundant ARs in the brain, may lack sufficient amounts of androgen to activate them. This seems highly probable because the treatment of female medaka and other teleosts with 11KT as adults readily induces male-typical behaviors, including courtship and aggression (9, 11, 12).

We found that *cyp19a1b*-deficient females are less receptive to male courtship, demonstrating that brain-derived estrogens influence female, as well as male, mating behavior. This finding is notable when considering that, contrary to mammals and birds, estrogens have been deemed unnecessary for female sexual receptivity in teleosts (49). This notion essentially stems from the observation in several teleost species that exogenous prostaglandin PGF2 α is sufficient to restore sexual receptivity in females deprived of ovarian secretions by ovariectomy (50–52). However, we recently challenged it by showing that female medaka deficient for *esr2b* are completely unreceptive to males, and thus estrogens play a critical role in female receptivity (12). Given that estrogen/Esr2b signaling feminizes the neural substrates that mediate mating behavior, while PGF2 α signaling triggers female sexual receptivity, the present finding provides a likely explanation for this apparent contradiction, namely, that brain-derived estrogens, rather than or in addition to ovarian-derived circulating estrogens, may function upstream of PGF2 α signaling to

mediate female receptivity. This scheme potentially applies to other teleost species because a reduction in female receptivity due to *cyp19a1b* deficiency has been also observed in zebrafish (37 [↗](#)).

We further found that *cyp19a1b*-deficient females court other females. To our knowledge, only a few neuronal genes have been identified whose depletion causes females to display male-typical behaviors in vertebrates. These include *trpc2* (encoding a cation channel crucial for olfactory cue-evoked signaling in the vomeronasal organ) and *dmnt3a* (encoding a methyltransferase responsible for de novo DNA methylation) in mice, and *esr2b* in medaka (12 [↗](#), 53 [↗](#), 54 [↗](#)). Importantly, deficiency of *cyp19a1b*, unlike that of *esr2b*, did not completely preclude female-typical mating behavior in females, but caused them to exhibit both male- and female-typical mating behaviors. This observation suggests that, in teleosts, the neural circuits mediating these behaviors coexist in the female brain and may be activated simultaneously in the absence of brain estrogens. This probably also holds true for the male brain, given that *cyp19a1b*-deficient males also showed impaired, but not completely abolished, male-typical behaviors and were actively courted by *esr2b*-deficient females. It is thus plausible that brain-derived estrogens are required to prevent the simultaneous activation of neural circuits mediating male- and female-typical behaviors, thereby contributing to the mutually exclusive presentation of these behaviors. Such effects of brain estrogens are likely to be particularly relevant in teleosts, where many species spontaneously undergo sex reversal and their sexual phenotypes, behavioral or otherwise, are highly labile throughout life (55 [↗](#), 56 [↗](#)). This may account for the extremely high levels of aromatase activity in the teleost brain.

In summary, we have shown that brain-derived estrogens promote male-typical behaviors by increasing brain sensitivity to testicular androgens through the stimulation of AR expression. Our findings challenge the prevailing view that brain-derived estrogens have little effect on male-typical behaviors in species where testicular androgens elicit these behaviors directly without aromatization to estrogens in the brain and unveil a previously unappreciated mechanism of the action of brain-derived estrogens. Because these species account for the majority of vertebrates, with rodents and some birds being the only known exceptions, the mechanism of their action that we have identified in medaka may be evolutionarily ancient and widely conserved across species.

Materials and Methods

Animals and cell lines

Wild-type d-rR strain medaka and mutant medaka deficient for *cyp19a1b* and *esr2a* (generated in this study), *esr1* (36 [↗](#)), and *esr2b* (12 [↗](#)) were maintained in a recirculating system at 28°C on a 14/10-hour light/dark cycle. Three or four times a day, they were fed live brine shrimp and dry food (Otohime; Marubeni Nisshin Feed, Tokyo, Japan). Sexually mature adults (2–6 months) were used for experiments, and tissues were consistently sampled 1–5 hours after lights on. In each experiment, siblings raised under the same conditions were used as the comparison group to control for the effects of genetic diversity and environmental variation. All fish were handled in accordance with the guidelines of the Institutional Animal Care and Use Committee of the University of Tokyo.

HEK293T and HeLa cells used in this study were confirmed to be mycoplasma-free (Biotherapy Institute of Japan, Tokyo, Japan) and authenticated by short tandem repeat profiling (National Institute of Biomedical Innovation, Osaka, Japan).

Generation of mutant medaka

cyp19a1b-deficient medaka were generated essentially as previously described (12). In brief, a TILLING library of 5760 chemically mutagenized medaka (28) was screened for mutations in exons 3, 4, and 5 of *cyp19a1b* by direct sequencing of PCR-amplified fragments. A founder (ID: 57D05) with a nonsense mutation in exon 4 (K105*) was identified (Fig. S1, A and B) and backcrossed to the wild-type d-rR strain for more than six generations to eliminate background mutations. Heterozygous males and females were intercrossed to generate wild-type, heterozygous, and homozygous siblings. All experimental fish were subjected to genotyping by PCR amplification across the mutation, followed by high-resolution melting (HRM) analysis, using the primers and probe listed in Table S2. HRM analysis was performed on the LightCycler 480 System II (Roche Diagnostics, Basel, Switzerland) using the LCGreen Plus dye (BioFire Defense, Salt Lake City, UT).

esr2a-deficient medaka were generated by using the CRISPR/Cas9 system. A CRISPR RNA (crRNA) targeting exon 3 of *esr2a* (Fasmac, Kanagawa, Japan) (Fig. S5A) was injected with trans-activating crRNA (Fasmac) and Cas9 protein (Nippon Gene Co. Ltd., Tokyo, Japan) into the cytoplasm of one- or two-cell stage embryos. The resulting fish were screened for germline mutations by outcrossing to wild-type fish and testing progeny for target site mutations by T7 endonuclease I assay (57) and direct sequencing. Two founders were identified that reproducibly yielded progeny carrying frameshift mutations that eliminated the DNA- and ligand-binding domains of *Esr2a*: one yielded progeny carrying an 8-bp deletion ($\Delta 8$); the other yielded progeny carrying a 4-bp deletion ($\Delta 4$) (Fig. S5, A and B). The DNA- and ligand-binding domains of medaka *Esr2a* were identified by sequence alignment with yellow perch (*Perca flavescens*) *Esr2a*, for which these domain locations have been reported (58). The progeny were intercrossed to obtain wild-type, heterozygous, and homozygous siblings. The genotype of each experimental fish was determined by HRM analysis as described above.

A previous study reported that *esr2a*-deficient female medaka cannot release eggs due to oviduct atresia (59). Likewise, some *esr2a*-deficient females generated in this study, despite the limited sample size, exhibited spawning behavior but were unable to release eggs ($\Delta 8$ line: 2/3; $\Delta 4$ line: 1/1), while such failure was not observed in wild-type females ($\Delta 8$ line: 0/12; $\Delta 4$ line: 0/11). These results support the effective loss of *esr2a* function.

Measurement of sex steroid levels

Brain and peripheral levels of sex steroids were determined by enzyme-linked immunosorbent assay (ELISA) (12). Total lipids were extracted from the brain and peripheral tissues (from the caudal half) of *cyp19a1b*^{+/+}, *cyp19a1b*^{+/-}, and *cyp19a1b*^{-/-} males and females with chloroform/methanol (2:1, v/v) and purified on a Sep-Pak C18 Plus Light Cartridge (Waters Corporation, Milford, MA). Tissue levels of E2, testosterone, and 11KT were measured by using Estradiol, Testosterone, and 11-Keto Testosterone ELISA kits, respectively (Cayman Chemical Company, Ann Arbor, MI).

Mating behavior test

Mating behavior was tested essentially as previously described (39). In brief, on the day before behavioral testing, each focal fish (*cyp19a1b*-, *esr1*-, and *esr2a*-deficient lines) was placed with a stimulus fish (wild-type females in the tests shown in Fig. 1, C and D, Fig. 6, E and F, Fig. S7, and Fig. S8, A and B; *esr2b*-deficient females in Fig. 1, E to H, Fig. 2, A to E, Fig. 5, E to J, and Fig. S8, C and D; wild-type males in Fig. 6, A to C) in a 2-liter rectangular tank with a perforated transparent partition separating them. The set-up was modified for E2-treated males, which were kept on E2 treatment and not introduced to the test tanks until the day of testing to ensure the efficacy of E2 treatment. The partition was removed 1 hour after lights on, and fish

were allowed to interact for 30 min while their behavior was recorded with a digital video camera (HC-V360MS, HC-VX985M, or HC-W870M; Panasonic Corporation, Osaka, Japan). The first 15 min of the recording was used to calculate the latency to the first following, courtship display, wrapping, and spawning. The period of interaction/recording was extended to 2 hours in tests of courtship displays received from the stimulus *esr2b*-deficient female and in tests of mating behavior between females, because they take longer to initiate courtship (12 [DOI](#)). In tests using an *esr2b*-deficient female as the stimulus fish, where the latency to spawn could not be calculated because these fish were unreceptive to males and did not spawn, the sexual motivation of the focal fish was instead assessed by counting the number of courtship displays and wrapping attempts in 30 min. The number of these mating acts was also counted in tests to evaluate the receptivity of females. In tests of mating behavior between two females, the stimulus female was marked with a small notch in the caudal fin to distinguish it from the focal female.

Aggressive behavior test

To test male–male aggressive behavior (29 [DOI](#)), four males of the same genotype (*cyp19a1b*-, *esr1*-, and *esr2a*-deficient lines) that were unfamiliar with one another were placed together in a test tank 1 hour after lights on. After 1 min of acclimation to the tank, fish were allowed to interact for 15 min while their behavior was video-recorded as described above. The total number of each aggressive act (chase, fin display, circle, strike, and bite) displayed by the four males in the tank was counted manually from the video recordings.

E2 treatment

cyp19a1b^{+/+} and *cyp19a1b*^{-/-} males were treated with 1 ng/ml of E2 (Fujifilm Wako Pure Chemical, Osaka, Japan), which was first dissolved in 100% ethanol (vehicle), or with the vehicle alone by immersion in water for 4 days, with daily water changes to maintain the nominal concentration. The mating and aggressive behaviors of males before and after E2 treatment were evaluated as described above. The expression of *ara* and *arb* in the brain of E2- and vehicle-treated males was investigated by single-label in situ hybridization as described below. Although the exact increase in brain E2 levels following E2 treatment was not quantified, the observed positive effects on behavior and gene expression suggest that it was sufficient.

Single-label in situ hybridization

Digoxigenin (DIG)-labeled cRNA probes were generated by in vitro transcription using DIG RNA Labeling Mix (Roche Diagnostics), T7 RNA polymerase (Roche Diagnostics), and the following cDNA fragments: *ara*, nucleotide 16–1030 of GenBank NM_001122911 (1015 bp); *arb*, 53–1233 of NM_001104681 (1181 bp); *vt*, 1–845 of NM_001278891 (845 bp); *gal*, 5–533 of LC532140 (529 bp); *npba*, 1–645 of NM_001308979 (645 bp); and *esr2b*, 2080–3288 of XM_020713365 (1209 bp).

Single-label in situ hybridization was essentially done as described (60 [DOI](#)). In brief, brains dissected from males and females of the *cyp19a1b*-deficient line (analysis of *ara*, *arb*, *vt*, *gal*, *npba*, and *esr2b*) and males of the *esr1*-, *esr2a*-, and *esr2b*-deficient lines (analysis of *ara* and *arb*) were fixed in 4% paraformaldehyde and embedded in paraffin. Serial coronal sections of 10-μm thickness were hybridized with a DIG-labeled probe. Hybridization signal was detected with anti-DIG conjugated to alkaline phosphatase (RRID: AB_514497; Roche Diagnostics) and visualized using 5-bromo-4-chloro-3-indolyl phosphate/nitro blue tetrazolium substrate (Roche Diagnostics). After color development for 15 min (*gal*), 40 min (*npba*), 2 hours (*vt*), or overnight (*ara*, *arb*, and *esr2b*), sections were imaged using a VS120 virtual slide microscope (Olympus, Tokyo, Japan). The total area of signal across all relevant sections, including both hemispheres, was calculated for each brain nucleus using Olyvia software (Olympus). Images were converted to a 256-level intensity scale, and pixels with intensities from 161 to 256 were considered signals. All sections used for comparison were processed in the same batch, without corrections between samples. Medaka brain atlases (61 [DOI](#), 62 [DOI](#)) were used to identify brain nuclei.

Transcriptional activity assay

Each full-length coding region of medaka *Esr1* (GenBank XM_020714493), *Esr2a* (NM_001104702), and *Esr2b* (XM_020713365) cDNA was amplified by PCR and inserted into the expression vector pcDNA3.1/V5-His-TOPO (Thermo Fisher Scientific, Waltham, MA). Each nucleotide sequence of the 5'-flanking region of *ara* and *arb* was retrieved from the Ensembl medaka genome assembly and analyzed for potential canonical ERE-like sequences using Jaspar (version 5.0_alpha) and Match (public version 1.0) with default settings. No ERE-like sequence was detected in *ara*; therefore, the gene body and 3'-flanking region were also analyzed in this case. The transcription initiation site for *ara* and *arb* was determined based on an expressed sequence tag clone deposited in National BioResource Project (NBRP) Medaka (olo36n18) and GenBank NM_001104681, respectively. Medaka bacterial artificial chromosome (BAC) clones containing the *ara* (clone ID: ola1-111G01) and *arb* (ola1-192H15) loci were obtained from NBRP Medaka. A 4530-bp genomic DNA fragment upstream of *ara* exon 3 (comprising 2330 bp of the 5'-flanking region, exon 1, intron 1, exon 2, intron 2, and the first 32 bp of exon 3) was amplified by PCR from the BAC clone, fused to a P2A self-cleaving peptide sequence (63 [aa](#)) at the 3' end, and inserted into the *NheI* site of the luciferase reporter vector pGL4.10 (Promega, Madison, WI). Similarly, a 3995-bp fragment upstream of the first methionine codon of *arb* (comprising 3815 bp of the 5'-flanking region and 180 bp of exon1) was PCR-amplified and inserted into pGL4.10. The resulting constructs were transiently transfected into HEK293T (*ara*) or HeLa (*arb*) cells, together with an internal control vector pGL4.74 (Promega) and the *Esr1*, *Esr2a*, or *Esr2b* expression construct using Lipofectamine LTX (Thermo Fisher Scientific). Six hours after transfection, cells were incubated for 18 hours without and with E2 at doses of 10^{-11} M, 10^{-9} M, and 10^{-7} M in Dulbecco's modified Eagle's medium (phenol red-free) containing 5% charcoal/dextran-stripped fetal bovine serum (Cytiva, Marlborough, MA). Luciferase activity was determined using the Dual-Luciferase Reporter Assay System (Promega) on the GloMax 20/20n Luminometer (Promega). All assays were conducted in duplicate or triplicate and repeated independently three times.

Further assays were performed with luciferase reporter constructs carrying point mutations in the ERE-like sequences to identify the ERE responsible for E2 induction of *ara* and *arb* transcription. Each half-site of the responsible ERE-like sequence was mutated into a *HindIII* recognition site using the PrimeSTAR Mutagenesis Basal Kit (Takara Bio, Shiga, Japan). Transcriptional activity assays with these constructs were done as described above, except that a single dose of E2 (10^{-7} M) was used.

Double-label in situ hybridization

DIG-labeled cRNA probes for *ara* and *arb* were prepared as described above. Fluorescein-labeled cRNA probes for *esr1* and *esr2a* were generated by in vitro transcription using Fluorescein RNA Labeling Mix (Roche Diagnostics), SP6 or T7 RNA polymerase (Roche Diagnostics), and the following cDNA fragments: *esr1*, nucleotides 1694–2781 of XM_020714493 (1088 bp); *esr2a*, 1838–2276 of NM_001104702 plus 763 bp of 3'-untranslated sequence derived from the Ensembl medaka genome assembly (439 bp).

Double-label in situ hybridization was done as described previously (36 [aa](#)), with minor modifications. In brief, brains dissected from wild-type males were fixed in 4% paraformaldehyde, embedded in paraffin, and coronally sectioned at 10- μ m thickness. Sections were hybridized simultaneously with the DIG-labeled *ara* or *arb* probe and fluorescein-labeled *esr1* or *esr2a* probe. DIG was detected with a mouse anti-DIG antibody (RRID: AB_304362; Abcam, Cambridge, UK) and visualized using the Alexa Fluor 555 Tyramide SuperBoost Kit, goat anti-mouse IgG (Thermo Fisher Scientific); fluorescein was detected with an anti-fluorescein antibody conjugated to horseradish peroxidase (RRID: AB_2737388; PerkinElmer, Waltham, MA) and visualized using the TSA Plus Fluorescein System (PerkinElmer). Cell nuclei were counterstained with DAPI. Fluorescent images were acquired with a Leica TCS SP8 confocal laser scanning microscope (Leica Microsystems,

Wetzlar, Germany) and the following excitation/emission wavelengths: 552/620–700 nm (Alexa Fluor 555), 488/495–545 nm (fluorescein), and 405/410–480 nm (DAPI). Cells were identified as coexpressing the two genes when Alexa Fluor 555 and fluorescein signals were clearly observed in the cytoplasm surrounding DAPI-stained nuclei, with intensities markedly stronger than the background noise.

Measurement of spatial distances between fish

The distance between the two fish in the test tank was measured as described elsewhere (12). On the day preceding measurement, each focal female (*cyp19a1b*-deficient line) and a stimulus wild-type male were placed together in a 2-liter rectangular tank with a perforated transparent partition separating them. On the following day after lights on, the fish were transferred to a 25-cm (i.d.) cylindrical test tank containing 1.8 L of dechlorinated tap water with a transparent partition separating them. One hour after lights on, the partition was removed and the behavior of the fish was video-recorded for 10 min. Distances between fish were calculated by using UMA Tracker (64).

Statistical analysis

All quantitative data were expressed as mean $\pm$ standard error of the mean (SEM). On graphs, individual data points were plotted to indicate the underlying distribution. Behavioral time-series data were analyzed using Kaplan–Meier plots with the inclusion of fish that did not exhibit the given act within the test period.

Statistical analyses were done using GraphPad Prism (GraphPad Software, San Diego, CA). Data between two groups were compared using unpaired two-tailed Student's *t* test. Welch's correction was applied if the *F* test indicated that the variance between groups was significantly different. The Bonferroni–Dunn correction was applied for multiple comparisons between two groups. To compare data among more than two groups, one-way analysis of variance (ANOVA) was performed, followed by either Bonferroni's (comparisons among experimental groups) or Dunnett's (comparisons between untreated and E2-treated groups in Fig. 4, C and D) post hoc test. If the Brown-Forsythe test indicated a significant difference in variance across groups, the data were instead analyzed using the non-parametric Kruskal–Wallis test, followed by Dunn's post hoc test. Differences between Kaplan–Meier curves were tested for significance using the Gehan–Breslow–Wilcoxon test (with Bonferroni's correction for more than two comparisons). Fish that spawned without any courtship display were excluded from the analysis of courtship display because it was not appropriate to treat them either as fish that did not perform courtship displays within the test duration or as fish that performed the first courtship display 0 seconds before spawning. Specifically, 7/18 *cyp19a1b*^{+/+}, 11/18 *cyp19a1b*^{+/-}, and 6/18 *cyp19a1b*^{-/-} males were excluded in Fig. 1D; 6/10 *cyp19a1b*^{+/+}, 3/10 *cyp19a1b*^{+/-}, and 6/10 *cyp19a1b*^{-/-} females were excluded in Fig. 6B; 2/23 *esr1*^{+/+} and 5/24 *esr1*^{-/-} males were excluded in Fig. S7; 2/24 *esr2a*^{+/+} and 3/23 *esr2a*^{-/-} males were excluded in Fig. S8A; 0/23 *esr2a*^{+/+} and 0/23 *esr2a*^{-/-} males were excluded in Fig. S8B. All data points were included in the analyses and no outliers were defined.

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Additional files

Supplemental Tabels and Figures [↗](#)

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Reviewer #1 (Public review):

Summary:

This research group has consistently performed cutting-edge research aiming to understand the role of hormones in the control of social behaviors, specifically by utilizing the genetically-tractable teleost fish, medaka, and the current work is no exception. The overall claim they make, that estrogens modulate social behaviors in males and females is supported, with important caveats. For one, there is no evidence these estrogens are generated by "neurons" as would be assumed by their main claim that it is NEUROestrogens that drive this effect. While indeed the aromatase they have investigated is expressed solely in the brain, in most teleosts, brain aromatase is only present in glial cells (astrocytes, radial glia). The authors should change this description so as not to mislead the reader. Below I detail more specific strengths and weaknesses of this manuscript.

Strengths:

- Excellent use of the medaka model to disentangle the control of social behavior by sex steroid hormones
- The findings are strong for the most part because deficits in the mutants are restored by the molecule (estrogens) that was no longer present due to the mutation
- Presentation of the approach and findings are clear, allowing the reader to make their own inferences and compare them with the authors'
- Includes multiple follow-up experiments, which leads to tests of internal replication and an impactful mechanistic proposal
- Findings are provocative not just for teleost researchers, but for other species since, as the authors point out, the data suggest mechanisms of estrogenic control of social behaviors may be evolutionary ancient

Weaknesses:

- As stated in the summary, the authors are attributing the estrogen source to neurons and there isn't evidence this is the case. The impact of the findings doesn't rest on this either
- The d4 versus d8 *esr2a* mutants showed different results for aggression. The meaning and implications of this finding are not discussed, leaving the reader wondering
- Lack of attribution of previous published work from other research groups that would provide the proper context of the present study
- There are a surprising number of citations not included; some of the ones not included argue against the authors' claims that their findings were "contrary to expectation"
- The experimental design for studying aggression in males has flaws. A standard test like a resident-intruder test should be used.
- While they investigate males and females, there are fewer experiments and explanations for the female results, making it feel like a small addition or an aside
- The statistics comparing "experimental to experimental" and "control to experimental" isn't appropriate

<https://doi.org/10.7554/eLife.97106.2.sa2>

Reviewer #3 (Public review):**Summary:**

Taking advantage of the existence in fish of two genes coding for estrogen synthase, the enzyme aromatase, one mostly expressed in the brain (Cyp19a1b) and the other mostly found in the gonads (Cyp19a1a), this study investigates the role of brain-derived estrogens in the control of sexual and aggressive behavior in medaka. The constitutive deletion of Cyp19a1b markedly reduced brain estrogen content in males and to a lesser extent in females. These effects are accompanied by reduced sexual and aggressive behavior in males and reduced preference for males in females. These effects are reversed by adult treatment with supporting a role for estrogens. The deletion of Cyp19a1b is associated with a reduced expression of the genes coding for the two androgen receptors, *ara* and *arb*, in brain regions involved in the regulation of social behavior. The analysis of the gene expression and behavior of mutants of estrogen receptors indicates that these effects are likely mediated by the activation of the *esr1* and *esr2a* isoforms. These results provide valuable insight into the role of estrogens in social behavior in the most abundant vertebrate taxon, however the conclusion of brain-derived estrogens awaits definitive confirmation.

Strengths:

- Evaluation of the role of brain "specific" Cyp19a1 in male teleost fish, which as a taxon are more abundant and yet proportionally less studied than the most common birds and rodents. Therefore, evaluating the generalizability of results from higher vertebrates is important. This approach also offers great potential to study the role of brain estrogen production in females, an understudied question in all taxa.
- Results obtained from multiple mutant lines converge to show that estrogen signaling, likely synthesized in the brain drives aspects of male sexual behavior.
- The comparative discussion of the age-dependent abundance of brain aromatase in fish vs mammals and its role in organization vs activation is important beyond the study of the targeted species.
- The authors have made important corrections to tone down some of the conclusions which are more in line with the results.

Weaknesses:

- No evaluation of the mRNA and protein products of Cyp19a1b and ESR2a are presented, such that there is no proper demonstration that the mutation indeed leads to aromatase reduction. The conclusion that these effects dependent on brain derived estrogens is therefore only supported by measures of E2 with an EIA kit that is not validated. No discussion of these shortcomings is provided in the discussion thus further weakening the conclusion manuscript.
- Most experiments are weakly powered (low sample size).
- The variability of the mRNA content for a same target gene between experiments (genotype comparison vs E2 treatment comparison) raises questions about the reproducibility of the data (apparent disappearance of genotype effect).

Conclusions:

Overall, the claims regarding role of estrogens originating in the brain on male sexual behavior is supported by converging evidence from multiple mutant lines. The role of brain-derived estrogens on gene expression in the brain is weaker as are the results in females.

<https://doi.org/10.7554/eLife.97106.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public Review)>

Summary:

This research group has consistently performed cutting-edge research aiming to understand the role of hormones in the control of social behaviors, specifically by utilizing the genetically tractable teleost fish, medaka, and the current work is no exception. The overall claim they make, that estrogens modulate social behaviors in males and females is supported, with important caveats. For one, there is no evidence these estrogens are generated by "neurons" as would be assumed by their main claim that it is NEUROestrogens that drive this effect. While indeed the aromatase they have investigated is expressed solely in the brain, in most teleosts, brain aromatase is only present in glial cells (astrocytes, radial glia). The authors should change this description so as not to mislead the reader. Below I detail more specific strengths and weaknesses of this manuscript.

We thank the reviewer for this very positive evaluation of our work and greatly appreciate their helpful comments and suggestions for improving the manuscript. We agree with the comment that the term “neuroestrogens” is misleading. Therefore, we have replaced “neuroestrogens” with “brain-derived estrogens” or “brain estrogens” throughout the manuscript, including the title.

In the following sections, “neuroestrogens” has been revised to align with the surrounding context.

Line 21: “in the brain, also known as neuroestrogens,” → “in the brain.”

Line 28: “neuroestrogens” → “these estrogens.”

Line 30: “mechanism of action of neuroestrogens” → “mode of action of brain-derived estrogens.”

Line 43: “brain-derived estrogens, also called neuroestrogens,” → “estrogens.”

Line 74: “neuroestrogen synthesis is selectively impaired while gonadal estrogen synthesis remains intact” → “estrogen synthesis in the brain is selectively impaired while that in the gonads remains intact.”

Line 77: “neuroestrogens” → “these estrogens.”

Line 335: “levels of neuroestrogens” → “brain estrogen levels.”

Line 338: “neuroestrogens” → “these estrogens.”

Line 351: “neuroestrogens” → “these estrogens.”

Line 357: “neuroestrogen action” → “the action of brain-derived estrogens.”

Line 359: “neuroestrogens” → “estrogen synthesis in the brain.”

Line 390: “active synthesis of neuroestrogens” → “active estrogen synthesis in the brain.”

Line 431: “neuroestrogens” → “estrogens in the brain.”

Line 431: “neuroestrogen action” → “the action of brain-derived estrogens.”

Line 433: “neuroestrogen action” → “their action.”

Strengths:

Excellent use of the medaka model to disentangle the control of social behavior by sex steroid hormones.

The findings are strong for the most part because deficits in the mutants are restored by the molecule (estrogens) that was no longer present due to the mutation.

Presentation of the approach and findings are clear, allowing the reader to make their own inferences and compare them with the authors'.

Includes multiple follow-up experiments, which lead to tests of internal replication and an impactful mechanistic proposal.

Findings are provocative not just for teleost researchers, but for other species since, as the authors point out, the data suggest mechanisms of estrogenic control of social behaviors may be evolutionarily ancient.

We again thank the reviewer for their positive evaluation of our work.

Weaknesses:

(1) As stated in the summary, the authors attribute the estrogen source to neurons and there isn't evidence this is the case. The impact of the findings doesn't rest on this either.

As noted in Response to reviewer #1's summary comment, we have replaced “neuroestrogens” with “brain-derived estrogens” or “brain estrogens” throughout the manuscript.

Line 63: We have also added the text “In teleost brains, including those of medaka, aromatase is exclusively localized in radial glial cells, in contrast to its neuronal localization in rodent brains (18–20).” Following this addition, “This observation suggests” in the subsequent sentence has been replaced with “These observations suggest.”

The following references (#18–20), cited in the newly added text above, have been included in the reference list, with other references renumbered accordingly:

P. M. Forlano, D. L. Deitcher, D. A. Myers, A. H. Bass, Anatomical distribution and cellular basis for high levels of aromatase activity in the brain of teleost fish: aromatase enzyme and mRNA expression identify glia as source. *J. Neurosci.* 21, 8943–8955 (2001).

N. Diotel, Y. Le Page, K. Mouriec, S. K. Tong, E. Pellegrini, C. Vaillant, I. Anglade, F. Brion, F. Pakdel, B. C. Chung, O. Kah, Aromatase in the brain of teleost fish: expression, regulation and putative functions. *Front. Neuroendocrinol.* 31, 172–192 (2010).

A. Takeuchi, K. Okubo, Post-proliferative immature radial glial cells female-specifically express aromatase in the medaka optic tectum. *PLoS One* 8, e73663 (2013).

(2) *The d4 versus d8 esr2a mutants showed different results for aggression. The meaning and implications of this finding are not discussed, leaving the reader wondering.*

Line 282: As the reviewer correctly noted, circles were significantly reduced in mutant males of the Δ8 line, whereas no significant reduction was observed in those of the Δ4 line. However, a tendency toward reduction was evident in the Δ4 line ($P = 0.1512$), and both lines showed significant differences in fin displays. Based on these findings, we believe our conclusion that *esr2a*^{-/-} males exhibit reduced aggression remains valid. To clarify this point and address potential reader concerns, we have revised the text as follows: “*esr2a*^{-/-} males from both the Δ8 and Δ4 lines exhibited significantly fewer fin displays than their wildtype siblings ($P = 0.0461$ and 0.0293 , respectively). Circles followed a similar pattern, with a significant reduction in the Δ8 line ($P = 0.0446$) and a comparable but non-significant decrease in the Δ4 line ($P = 0.1512$) (Fig. 5L; Fig. S8E), showing less aggression.”

(3) *Lack of attribution of previously published work from other research groups that would provide the proper context of the present study.*

In response to this and other comments from this reviewer, we have revised the Introduction and Discussion sections as follows.

Line 56: “solely responsible” in the Introduction has been modified to “largely responsible”.

Line 57: “This is consistent with the recent finding in medaka fish (*Oryzias latipes*) that estrogens act through the ESR subtype Esr2b to prevent females from engaging in male-typical courtship (10)” has been revised to “This is consistent with recent observations in a few teleost species that genetic ablation of AR severely impairs male-typical behaviors (13–16) and with findings in medaka fish (*Oryzias latipes*) that estrogens act through the ESR subtype Esr2b to prevent females from engaging in male-typical courtship (12)” to include previous studies on the behavior of AR mutant fish (Yong et al., 2017; Alward et al., 2020; Ogino et al., 2023; Nishiike and Okubo, 2024) in the Introduction.

Line 65: “It is worth mentioning that systemic administration of estrogens and an aromatase inhibitor increased and decreased male aggression, respectively, in several teleost species, potentially reflecting the behavioral effects of brain-derived estrogens (21–24)” has been added to the Introduction. This addition provides an overview of previous studies on the effects of estrogens and aromatase on male fish aggression (Hallgren et al., 2006; O’Connell and Hofmann, 2012; Huffman et al., 2013; Jalabert et al., 2015).

Line 367: “treatment of males with an aromatase inhibitor reduces their male-typical behaviors (31–33)” has been edited to read “treatment of males with an aromatase inhibitor reduces their male-typical behaviors, while estrogens exert the opposite effect (21–24).”

After the revisions described above, the following references (#13, 14, and 22) have been added to the reference list, with other references renumbered accordingly:

L. Yong, Z. Thet, Y. Zhu, Genetic editing of the androgen receptor contributes to impaired male courtship behavior in zebrafish. *J. Exp. Biol.* 220, 3017–3021 (2017).

B. A. Alward, V. A. Laud, C. J. Skalnik, R. A. York, S. A. Juntti, R. D. Fernald, Modular genetic control of social status in a cichlid fish. *Proc. Natl. Acad. Sci. U.S.A.* 117, 28167–28174 (2020).

L. A. O’Connell, H. A. Hofmann, Social status predicts how sex steroid receptors regulate complex behavior across levels of biological organization. *Endocrinology* 153, 1341–1351 (2012).

(4) There are a surprising number of citations not included; some of the ones not included argue against the authors' claims that their findings were "contrary to expectation".

Line 68: As detailed in Response to reviewer #1's comment 3 on weaknesses, we have cited previous studies on the effects of estrogens and aromatase on male fish aggression (Hallgren et al., 2006; O'Connell and Hofmann, 2012; Huffman et al., 2013; Jalabert et al., 2015) in the Introduction.

The following revisions have also been made to avoid phrases such as "contrary to expectation" and "unexpected."

Line 76: "Contrary to our expectations" → "Remarkably."

Line 109: "Contrary to this expectation, however" → "Nevertheless."

Line 135: "Again, contrary to our expectation, *cyp19a1b*^{-/-} males" → "*cyp19a1b*^{-/-} males."

Line 333: "unexpected" → "noteworthy."

Line 337: "unexpected" → "notable."

(5) The experimental design for studying aggression in males has flaws. A standard test like a resident intruder test should be used.

We agree that the resident-intruder test is the most commonly used method for assessing aggression. However, medaka form shoals and lack strong territoriality, and even slight dominance differences between the resident and the intruder can increase variability in the results, compromising data consistency. Therefore, in this study, we adopted an alternative approach: placing four unfamiliar males together in a tank and quantifying aggressive interactions in total. This method allows for the assessment of aggression regardless of territorial tendencies, making it more appropriate for our investigation.

(6) While they investigate males and females, there are fewer experiments and explanations for the female results, making it feel like a small addition or an aside.

We agree that the data and discussion for females are less extensive than for males. However, we have previously elucidated the mechanism by which estrogen/Esr2b signaling promotes female mating behavior (Nishiike et al., 2021, Curr Biol, 1699–1710). Accordingly, it follows that the new insights into female behavior gained from the *cyp19a1b* knockout model are more limited than those for males. Nevertheless, when combined with our prior findings, the female data in this study offer valuable insights, and the overall mechanism through which estrogens promote female mating behavior is becoming clearer. Therefore, we do not consider the female data in this study to be incomplete or merely supplementary.

(7) The statistics comparing "experimental to experimental" and "control to experimental" aren't appropriate.

The reviewer raises concerns about the statistical analysis used for Figures 4C and 4E, suggesting that Bonferroni's test should be used instead of Dunnett's test. However, Dunnett's test is commonly used to compare treatment groups to a reference group that receives no treatment, as in our study. Since we do not compare the treated groups with each other, we believe Dunnett's test is the most appropriate choice.

Line 619: The reviewer's concern may have arisen from the phrase "comparisons between control and experimental groups" in the Materials and Methods. We have revised it to "comparisons between untreated and E2-treated groups in Fig. 4, C and D" for clarity.

Reviewer #2 (Public Review):

Summary:

*The novelty of this study stems from the observations that neuro-estrogens appear to interact with brain androgen receptors to support male-typical behaviors. The study provides a step forward in clarifying the somewhat contradictory findings that, in teleosts and unlike other vertebrates, androgens regulate male-typical behaviors without requiring aromatization, but at the same time estrogens appear to also be involved in regulating male-typical behaviors. They manipulate the expression of one aromatase isoform, *cyp19a1b*, that is purported to be brain-specific in teleosts. Their findings are important in that brain estrogen content is sensitive to the brain-specific *cyp19a1b* deficiency, leading to alterations in both sexual behavior and aggressive behavior. Interestingly, these males have relatively intact fertility rates, despite the effects on the brain.*

We thank this reviewer for their positive evaluation of our work and constructive comments, which we found very helpful in improving the manuscript.

That said, the framing of the study, the relevant context, and several aspects of the methods and results raise concerns. Two interpretations need to be addressed/tempered:

*(1) that the rescue of *cyp19a1b* deficiency by tank-applied estradiol is not necessarily a brain/neuroestrogen mode of action, and*

Line 155: *cyp19a1b*-deficient males exhibited a severe reduction in brain E2 levels, yet their peripheral E2 levels remained comparable to those in wild-type males. Given this hormonal milieu and the lack of behavioral change in wild-type males following E2 treatment, the observed recovery of mating behavior in *cyp19a1b*-deficient males following E2 treatment can be best explained by the restoration of brain E2 levels. However, as the reviewer pointed out, we cannot rule out the possibility that bath-immersed E2 influenced behavior through an indirect peripheral mechanism. To address this concern, we have modified the text as follows: "These results suggest that reduced E2 in the brain is the primary cause of the mating defects, highlighting a pivotal role of brain-derived estrogens in male mating behavior. However, caution is warranted, as an indirect peripheral effect of bath-immersed E2 on behavior cannot be ruled out, although this is unlikely given the comparable peripheral E2 levels in *cyp19a1b*-deficient and wild-type males. In contrast to mating."

*(2) the large increases in peripheral and brain androgen levels in the *cyp19a1b* deficient animals imply some indirect/compensatory effects of lifelong *cyp19a1b* deficiency.*

As stated in line 151, androgen/AR signaling has a strong facilitative effect on male-typical behaviors in teleosts. If increased androgen levels in the periphery and brain affected behavior, the expected effect would be facilitative. However, *cyp19a1b*-deficient males exhibited impaired male-typical behaviors, suggesting that elevated androgen levels were unlikely to be responsible. Although chronic androgen elevation could cause androgen receptor desensitization, which could lead to behavioral suppression, our long-term androgen treatments have consistently promoted, rather than inhibited, male-typical behaviors (e.g., Nishiike et al., Proc Natl Acad Sci USA 121:e2316459121). Hence, this possibility is also highly unlikely.

Reviewer #3 (Public Review):

Summary:

Taking advantage of the existence in fish of two genes coding for estrogen synthase, the enzyme aromatase, one mostly expressed in the brain (Cyp19a1b) and the other mostly found in the gonads (Cyp19a1a), this study investigates the role of neuro-estrogens in the control of sexual and aggressive behavior in teleost fish. The constitutive deletion of Cyp19a1b reduced brain estrogen content by 87% in males and about 50% in females. It led to reduced sexual and aggressive behavior in males and reduced sexual behavior in females. These effects are reversed by adult treatment with estradiol thus indicating that they are activational in nature. The deletion of Cyp19a1b is associated with a reduced expression of the genes coding for the two androgen receptors, ara, and arb, in brain regions involved in the regulation of social behavior. The analysis of the gene expression and behavior of mutants of estrogen receptors indicates that these effects are likely mediated by the activation of the esr1 and esr2a isoforms. These results provide valuable insight into the role of neuro-estrogens in social behavior in the most abundant vertebrate taxa. While estrogens are involved in the organization of the brain and behavior of some birds and rodents, neuro-estrogens appear to play an activational role in fish through a facilitatory action of androgen signaling.

We thank this reviewer for their positive evaluation of our work and comments that have improved the manuscript.

Strengths:

Evaluation of the role of brain "specific" Cyp19a1 in male teleost fish, which as a taxa are more abundant and yet proportionally less studied than the most common birds and rodents. Therefore, evaluating the generalizability of results from higher vertebrates is important. This approach also offers great potential to study the role of brain estrogen production in females, an understudied question in all taxa.

Results obtained from multiple mutant lines converge to show that estrogen signaling drives aspects of male sexual behavior.

The comparative discussion of the age-dependent abundance of brain aromatase in fish vs mammals and its role in organization vs activation is important beyond the study of the targeted species.

We again thank the reviewer for their positive evaluation of our work.

Weaknesses:

(1) The new transgenic lines are under-characterized. There is no evaluation of the mRNA and protein products of Cyp19a1b and ESR2a.

We did not directly assess the function of *cyp19a1b* and *esr2a* in our mutant fish. However, the observed reduction in brain E2 levels, with no change in peripheral E2 levels, in *cyp19a1b*-deficient fish strongly supports the loss of *cyp19a1b* function. This is stated in the Results section (line 97) as follows: "These results show that *cyp19a1b*-deficient fish have reduced estrogen levels coupled with increased androgen levels in the brain, confirming the loss of *cyp19a1b* function."

Line 473: A previous study reported that female medaka lacking *esr2a* fail to release eggs due to oviduct atresia (Kayo et al., 2019, Sci Rep 9:8868). Similarly, in this study, some *esr2a*-

deficient females exhibited spawning behavior but were unable to release eggs, although the sample size was limited ($\Delta 8$ line: 2/3; $\Delta 4$ line: 1/1). In contrast, this was not observed in wild-type females ($\Delta 8$ line: 0/12; $\Delta 4$ line: 0/11). These results support the effective loss of *esr2a* function. To incorporate this information into the manuscript, the following text has been added to the Materials and Methods: “A previous study reported that *esr2a*-deficient female medaka cannot release eggs due to oviduct atresia (59). Likewise, some *esr2a*-deficient females generated in this study, despite the limited sample size, exhibited spawning behavior but were unable to release eggs ($\Delta 8$ line: 2/3; $\Delta 4$ line: 1/1), while such failure was not observed in wild-type females ($\Delta 8$ line: 0/12; $\Delta 4$ line: 0/11). These results support the effective loss of *esr2a* function.”

The following reference (#59), cited in the newly added text above, have been included in the reference list:

D. Kayo, B. Zempo, S. Tomihara, Y. Oka, S. Kanda, Gene knockout analysis reveals essentiality of estrogen receptor $\beta 1$ (*Esr2a*) for female reproduction in medaka. *Sci. Rep.* 9, 8868 (2019).

(2) The stereotypic sequence of sexual behavior is poorly described, in particular, the part played by the two sexual partners, such that the conclusions are not easily understandable, notably with regards to the distinction between motivation and performance.

Line 103: To provide a more detailed description of medaka mating behavior, we have revised the text from “The mating behavior of medaka follows a stereotypical pattern, wherein a series of followings, courtship displays, and wrappings by the male leads to spawning” to “The mating behavior of medaka follows a stereotypical sequence. It begins with the male approaching and closely following the female (following). The male then performs a courtship display, rapidly swimming in a circular pattern in front of the female. If the female is receptive, the male grasps her with his fins (wrapping), culminating in the simultaneous release of eggs and sperm (spawning).”

(3) The behavior of females is only assessed from the perspective of the male, which raises questions about the interpretation of the reduced behavior of the males.

In medaka, female mating behavior is largely passive, except for rejecting courtship attempts and releasing eggs. Therefore, its analysis relies on measuring the latency to receive following, courtship displays, or wrappings from the male and the frequency of courtship rejection or wrapping refusal. We understand the reviewer’s perspective that *cyp19a1b*-deficient females might not be less receptive but instead less attractive to males, potentially leading to reduced male mating efforts. However, since these females are approached and followed by males at levels comparable to wild-type females, this possibility appears unlikely. Moreover, *cyp19a1b*-deficient females tend to avoid males and exhibit a slightly female-oriented sexual preference. While these traits are closely associated with reduced sexual receptivity, they do not readily align with reduced sexual attractiveness. Therefore, it is more plausible to conclude that these females have decreased receptivity rather than being less attractive to males.

(4) At no point do the authors seem to consider that a reduced behavior of one sex could result from a reduced sensory perception from this sex or a reduced attractivity or sensory communication from the other sex.

Line 112: As noted above, the impaired mating behavior of *cyp19a1b*-deficient females is unlikely to be due to reduced attractiveness to males. Similarly, mating behavior tests using *esr2b*-deficient females as stimulus females suggest that the impaired mating behavior of *cyp19a1b*-deficient males cannot be attributed to reduced attractiveness to females. However,

the possibility that their impaired mating behavior could be attributed to altered cognition or sexual preference cannot be ruled out. To reflect this in the manuscript, we have revised the text “, suggesting that they are less motivated to mate” to “. These results suggest that they are less motivated to mate, though an alternative interpretation that their cognition or sexual preference may be altered cannot be dismissed.”

(5) *Aspects of the methods are not detailed enough to allow proper evaluation of their quality or replication of the data.*

In response to this and other specific comments from this reviewer, we have revised the Materials and Methods section to include more detailed descriptions of the methods.

Line 469: The following text has been added to describe the method for domain identification in medaka *Esr2a*: “The DNA- and ligand-binding domains of medaka *Esr2a* were identified by sequence alignment with yellow perch (*Perca flavescens*) *Esr2a*, for which these domain locations have been reported (58).”

The following reference (#58), cited in the newly added text above, have been included in the reference list:

S. G. Lynn, W. J. Birge, B. S. Shepherd, Molecular characterization and sex-specific tissue expression of estrogen receptor α (*esr1*), estrogen receptor β (*esr2a*) and ovarian aromatase (*cyp19a1a*) in yellow perch (*Perca flavescens*). *Comp. Biochem. Physiol. B Biochem. Mol. Biol.* 149, 126–147 (2008).

Line 540: The text “, and the total area of signal in each brain nucleus was calculated using Olyvia software (Olympus)” has been revised to include additional details on the single ISH method as follows: “. The total area of signal across all relevant sections, including both hemispheres, was calculated for each brain nucleus using Olyvia software (Olympus). Images were converted to a 256-level intensity scale, and pixels with intensities from 161 to 256 were considered signals. All sections used for comparison were processed in the same batch, without corrections between samples.”

Line 596: The following text has been added to include additional details on the double ISH method: “Cells were identified as coexpressing the two genes when Alexa Fluor 555 and fluorescein signals were clearly observed in the cytoplasm surrounding DAPI-stained nuclei, with intensities markedly stronger than the background noise.”

(6) *It seems very dangerous to use the response to a mutant abnormal behavior (ESR2-KO females) as a test, given that it is not clear what is the cause of the disrupted behavior.*

esr2b-deficient females have fully developed ovaries, a normal sex steroid milieu, and sexual attractiveness to males comparable to wild-type females, yet they are completely unreceptive to male courtship (Nishiike et al., 2021, *Curr Biol*, 1699–1710). Although, as the reviewer noted, the detailed mechanisms underlying this phenotype remain unclear, it is evident that the loss of estrogen/*Esr2b* signaling in the brain severely impairs sexual receptivity. Therefore, using *esr2b*-deficient females as stimulus females in the mating behavior test eliminates the influence of female sexual receptivity and male attractiveness to females, enabling the exclusive assessment of male mating motivation. This rationale is already presented in the Results section (lines 116–120), and we believe this experimental design offers a robust framework for assessing male mating motivation.

Additionally, the mating behavior test with *esr2b*-deficient females complemented the test with wildtype females, and its results were not the sole basis for our discussion of the male

mating behavior phenotype. The results of both tests were largely concordant, and we believe that the conclusions drawn from them are highly reliable.

Meanwhile, in the test with *esr2b*-deficient females, *cyp19a1b*-deficient males were courted more frequently by these females than wild-type males. As the reviewer noted, this may suggest an anomaly in the test. Accordingly, we have confined our discussion to the possibility that “Perhaps *cyp19a1b*^{-/-} males are misidentified as females by *esr2b*-deficient females because they are reluctant to court or they exhibit some female-like behavior” (line 131).

(7) Most experiments are weakly powered (low sample size) and analyzed by multiple T-tests while 2 way ANOVA could have been used in several instances. No mention of T or F values, or degrees of freedom.

Histological analysis was conducted with a relatively small sample size, as our previous experience suggested that interindividual variability in the results would not be substantial. As significant differences were detected in many analyses, further increasing the sample size is unnecessary.

Although two-way ANOVA could be used instead of multiple T-tests for analyzing the data in Figures 4D, 4F, 6D, S4A, and S4B, we applied the Bonferroni–Dunn correction to control for multiple pairwise comparisons in multiple T-tests. As this comparison method is equivalent to the post hoc test following two-way ANOVA, the statistical results are identical regardless of whether T-tests or two-way ANOVA are used.

For the data in Figures 4D, 4F, S4A, and S4B, the primary focus is on whether relative luciferase activity differs between E2-treated and untreated conditions for each mutant construct. Therefore, two-way ANOVA is not particularly relevant, as assessing the main effect of construct type or its interaction with E2 treatment does not provide meaningful insights. Similarly, in Figure 6D, the focus is solely on whether wild-type and mutant females differ in time spent at each distance. Given this, two-way ANOVA is unnecessary, as analyzing the main effect of distance is not meaningful.

Accordingly, two-way ANOVA was not employed in this study, and therefore, its corresponding F values were not included. As the figure legends specify the sample sizes for all analyses, specifying degrees of freedom separately was deemed unnecessary.

(8) The variability of the mRNA content for the same target gene between experiments (genotype comparison vs E2 treatment comparison) raises questions about the reproducibility of the data (apparent disappearance of genotype effect).

As the reviewer pointed out, the overall area of *ara* expression is larger in Figure 2J than in Figure 2F. However, the relative area ratios of *ara* expression among brain nuclei are consistent between the two figures, indicating the reproducibility of the results. Thus, this difference is unlikely to affect the conclusions of this study.

Additionally, the differences in *ara* expression in pPPp and *arb* expression in aPPp between wild-type and *cyp19a1b*-deficient males appear less pronounced in Figures 2J and 2K than in Figures 2F and 2H. This is likely attributable to the smaller sample size used in the experiments for Figures 2J and 2K, resulting in less distinct differences. However, as the same genotype-dependent trends are observed in both sets of figures, the conclusion that *ara* and *arb* expression is reduced in *cyp19a1b*-deficient male brains remains valid.

(9) The discussion confuses the effects of estrogens on sexual differentiation (developmental programming = permanent) and activation (= reversible activation of

brain circuits in adulthood) of the brain and behavior. Whether sex differences in the circuits underlying social behaviors exist is not clear.

We recognize that the effects of adult steroids are sometimes not considered to be sexual differentiation, as they do not differentiate the neural substrate, but rather transiently activate the already masculinized or feminized substrate. Arnold (2017, J Neurosci Res 95:291–300) contends that all factors that cause sex differences, including the transient effects of adult steroids, should be incorporated into a theory of sexual differentiation, and indeed, these effects may be the most potent proximate factors that make males and females different. We concur with this perspective and have adopted it as a foundation for our manuscript.

In teleosts, early developmental exposure to steroids has minimal impact, and sexual differentiation relies primarily on steroid action in adulthood (Okubo et al., 2022, Spectrum of Sex, pp. 111–133). This is evidenced by the effective reversal of sex-typical behaviors through experimental hormonal manipulation in adult teleosts and the absence of transient early-life steroid surges observed in mammals and birds. Accordingly, our discussion on brain sexual differentiation, including the statement in line 347, “This variation among species may represent the activation of neuroestrogen synthesis at life stages critical for sexual differentiation of behavior that are unique to each species”, remains well-supported. Additionally, given these considerations, while sex differences in neural circuit activation are evident in teleosts, substantial structural differences in these circuits are unlikely.

(10) Overall, the claims regarding the activational role of neuro-estrogens on male sexual behavior are supported by converging evidence from multiple mutant lines. The role of neuroestrogens on gene expression in the brain is mostly solid too. The data for females are comparatively weaker. Conclusions regarding sexual differentiation should be considered carefully.

We agree that the data for females are less extensive than for males. However, we have previously elucidated the mechanism by which estrogen/Esr2b signaling promotes female mating behavior (Nishiike et al., 2021). Accordingly, it follows that the new insights into female behavior gained from the *cyp19a1b* knockout model are more limited than those for males. Nevertheless, when integrated with our prior findings, the data on females in this study provide significant insights, and the overall mechanism through which estrogens promote female mating behavior is becoming clearer. Therefore, we do not consider the female data in this study to be incomplete or merely supplementary.

Recommendations For The Authors:

Reviewer #1 (Recommendations For The Authors):

*The authors set out to answer an intriguing question regarding the hormonal control of innate social behaviors in medaka. Specifically, they wanted to test the effects of *cyp19a1b* mutation on mating and aggression in males. They also test these effects in females. Their approach takes them down several distinct experimental pathways, including one investigating how *cyp19a1a* function is related to androgen receptor expression and how estrogens themselves may act on the androgen receptor to modulate its expression, as well as how different *esr* genes may be involved. The study and its results are valuable and a clear, general conclusion of a pathway from brain aromatase>estrogens>*esr* genes> androgen receptor can be made. This is important, novel, and impactful. However, there are issues with how the study logic is set up, the approach for assessing certain behaviors, the statistics used, the interpretation of findings, and placing the findings in the proper context based on previous work, which manifests as a general issue where previous work is not properly attributed to.*

Thank you for your thoughtful review. We have carefully addressed each specific comment, as detailed below.

Major comments:

(1) *The background for the rationale of the current study is misleading and lacks proper context. The authors root the logic of their experiment in determining whether estrogens regulate male-typical behaviors because the current assumption is androgens are "solely responsible" for male-typical behaviors in teleosts. This is not the case. Previous studies have shown aromatase/estrogens are involved in male-typical aggression in teleosts. For example, to name a couple:*

*Huffman, L. S., O'Connell, L. A., & Hofmann, H. A. (2013). Aromatase regulates aggression in the African cichlid fish *Astatotilapia burtoni*. *Physiology & behavior*, 112, 77-83.*

*O'Connell, L. A., & Hofmann, H. A. (2012). Social status predicts how sex steroid receptors regulate complex behavior across levels of biological organization. *Endocrinology*, 153(3), 1341-1351.*

And even a recent paper sheds light on a possible AR>aromatase.estradiol hypothesis of male typical behaviors:

*Lopez, M. S., & Alward, B. A. (2024). Androgen receptor deficiency is associated with reduced aromatase expression in the ventromedial hypothalamus of male cichlids. *Annals of the New York Academy of Sciences*.*

Interestingly, the authors cite Hufmann et al in the discussion, so I don't understand why they make the claims they do about estrogens and male-typical behavior.

Related to this, is an issue of proper attribution to published work. Indeed, missing are key references from lab groups using AR mutant teleosts. Here are a couple:

*Yong, L., Thet, Z., & Zhu, Y. (2017). Genetic editing of the androgen receptor contributes to impaired male courtship behavior in zebrafish. *Journal of Experimental Biology*, 220(17), 3017-3021.*

*Alward, B. A., Laud, V. A., Skalnik, C. J., York, R. A., Juntti, S. A., & Fernald, R. D. (2020). Modular genetic control of social status in a cichlid fish. *Proceedings of the National Academy of Sciences*, 117(45), 28167-28174.*

*Ogino, Y., Ansai, S., Watanabe, E., Yasugi, M., Katayama, Y., Sakamoto, H., ... & Iguchi, T. (2023). Evolutionary differentiation of androgen receptor is responsible for sexual characteristic development in a teleost fish. *Nature communications*, 14(1), 1428.*

As noted in Response to reviewer #1's comment 3 on weaknesses, we have revised the Introduction and Discussion sections as follows.

Line 56: "solely responsible" in the Introduction has been modified to "largely responsible".

Line 57: The text "This is consistent with the recent finding in medaka fish (*Oryzias latipes*) that estrogens act through the ESR subtype Esr2b to prevent females from engaging in male-typical courtship (10)" has been revised to "This is consistent with recent observations in a few teleost species that genetic ablation of AR severely impairs male-typical behaviors (13–16) and with findings in medaka fish (*Oryzias latipes*) that estrogens act through the ESR subtype Esr2b to prevent females from engaging in male-typical courtship (12)" to include previous studies on the behavior of AR mutant fish (Yong et al., 2017; Alward et al., 2020; Ogino et al., 2023; Nishiike and Okubo, 2024) in the Introduction.

Line 65: “It is worth mentioning that systemic administration of estrogens and an aromatase inhibitor increased and decreased male aggression, respectively, in several teleost species, potentially reflecting the behavioral effects of brain-derived estrogens (21–24)” has been added to the Introduction, providing an overview of previous studies on the effects of estrogens and aromatase on male fish aggression (Hallgren et al., 2006; O’Connell and Hofmann, 2012; Huffman et al., 2013; Jalabert et al., 2015).

Line 367: “treatment of males with an aromatase inhibitor reduces their male-typical behaviors (31– 33)” has been edited to read “treatment of males with an aromatase inhibitor reduces their male-typical behaviors, while estrogens exert the opposite effect (21–24).”

After the revisions described above, the following references (#13, 14, and 22) have been added to the reference list:

L. Yong, Z. Thet, Y. Zhu, Genetic editing of the androgen receptor contributes to impaired male courtship behavior in zebrafish. *J. Exp. Biol.* 220, 3017–3021 (2017).

B. A. Alward, V. A. Laud, C. J. Skalnik, R. A. York, S. A. Juntti, R. D. Fernald, Modular genetic control of social status in a cichlid fish. *Proc. Natl. Acad. Sci. U.S.A.* 117, 28167–28174 (2020).

L. A. O’Connell, H. A. Hofmann, Social status predicts how sex steroid receptors regulate complex behavior across levels of biological organization. *Endocrinology* 153, 1341–1351 (2012).

While Lopez and Alward (2024) provide valuable insights into the regulation of *cyp19a1b* expression by androgens, our study focuses specifically on the functional aspects of *cyp19a1b*. Expanding the discussion to include expression regulation would divert from the primary focus of our manuscript. For this reason, we have opted not to cite this reference.

(2) As it is now, the authors are only citing a book chapter/review from their own group. This is a serious issue as it does not provide the proper context for the work. The authors need to fix their issues of attribution to previously published work and the proper interpretation of the work that they are aware of as it pertains to ideas proposed on the roles of androgens and estrogens in the control of male-typical behaviors. This is also important to get the citations right because the common use of "contrary to expectations" when describing their results is actually not correct. Many of the observations are expected to a degree. However, this doesn't take away from a generally stellar experimental design and mostly clear results. The authors do not need to rely on enhancing the impact of their paper by making false claims of unexpected findings. The depth and clarity of your findings are where the impact of your work is.

As detailed in Response to reviewer #1’s comment 3 on weaknesses, we have cited previous studies on the effects of estrogens and aromatase on male fish aggression (Hallgren et al., 2006; O’Connell and Hofmann, 2012; Huffman et al., 2013; Jalabert et al., 2015) in the Introduction.

Additionally, as noted in Response to reviewer #1’s comment 4 on weaknesses, we have made the following revisions to avoid phrases such as “contrary to expectation” and “unexpected.”

Line 76: “Contrary to our expectations” → “Remarkably.”

Line 109: “Contrary to this expectation, however” → “Nevertheless.”

Line 135: “Again, contrary to our expectation, *cyp19a1b*^{-/-} males” → “*cyp19a1b*^{-/-} males.”

Line 333: “unexpected” → “noteworthy.”

Line 337: “unexpected” → “notable.”

(3) The experimental design for studying aggression in males has flaws. A standard test like a resident-intruder test should be used. An assay in which only male mutants are housed together? I do not understand the logic there and the logic for the approach isn't even explained. Too many confounds that are not controlled for. It makes it seem like an aspect of the study that was thrown in as an aside.

As noted in Response to reviewer #1's comment 5 on weaknesses, medaka form shoals and lack strong territoriality. As a result, even slight differences in dominance between the resident and intruder can substantially impact the outcomes of the resident-intruder test. Therefore, we adopted an alternative approach in this study.

(4) Hormonal differences in the mutants seem to vary based on sex, and the meaning of these differences, or how they affect interpreting the findings, wasn't discussed. There was no acknowledgment of the fact that female central E2 was still at 50%, meaning the "rescue" experiments using peripheral injections are not given the proper context. For example, this is different than giving a fish with only 16% of their normal central E2 an E2 injection. Missing as well is a clear hypothesis for why E2 injections did not rescue aggression deficits in cyp19a1b mutant males.

Line 385: As the reviewer pointed out, the degree of brain estrogen reduction in *cyp19a1b*-deficient fish differs greatly between males and females. This is likely because females receive a large supply of estrogens from the ovaries. Given that estrogen levels in *cyp19a1b*-deficient females were 50% of those in wild-type females, it can be inferred that half of their brain estrogens are synthesized locally, while the other half originates from the ovaries. This is an important finding, and we have already noted in the Discussion that “females have higher brain levels of estrogens, half of which are synthesized locally in the brain (i.e., neuroestrogens)” However, as this explanation was not sufficiently clear, we have revised it to “females have higher brain levels of estrogens, with half being synthesized locally and the other half supplied by the ovaries.”

The reviewer raised a concern that conducting the estrogen rescue experiment in females, where 50% of brain estrogens remain, might be inappropriate. However, as this experiment was conducted exclusively in males, this concern is not applicable.

Line 377: As noted in the reviewer's subsequent comment, the failure of aggression recovery in E2-treated *cyp19a1b*-deficient males could be due to insufficient induction of *ara/arb* expression in aggression-relevant brain regions. To address this concern, we have inserted the following statement into the Discussion after “the development of male behaviors may require moderate neuroestrogen levels that are sufficient to induce the expression of *ara* and *arb*, but not *esr2b*, in the underlying neural circuitry”: “This may account for the lack of aggression recovery in E2-treated *cyp19a1b*-deficient males in this study.”

(5) In relation to that, the "null" results may have some of the most interesting implications, but they are barely discussed. For example, what does it mean that E2 didn't restore aggression in male cyp19 mutants? Is this a brain region factor? Could this relate to findings from Lopez et al NYAS, where male and female Ara mutants show different effects on brain-region-specific aromatase expression? And maybe this relates to the different impact of estrogens on ar expression. Were the different effects impacted in aggression areas? Maybe this is why E2 injection didn't restore aggression in males. You could make the argument that: (1) E2 doesn't restore ar expression in aggression regions and that's why there was no rescue. Or (2) that the circuits in adulthood that regulate aggression are NOT dependent on aggression but in early development they

are. Another null finding not expanded on is why the two esr2a mutant lines showed differences. There is no reason to trust one line over the other, meaning we still don't know whether esr2a is required for latency to follow.

As stated in our response to the previous comment, we have added the following text to the Discussion (line 377): “This may account for the lack of aggression recovery in E2-treated *cyp19a1b*-deficient males in this study.” Meanwhile, as discussed in lines 341–342, it is highly unlikely that the neural circuits regulating aggression are primarily influenced by early-life estrogen exposure, because androgen administration in adulthood alone is sufficient to induce high levels of aggression in both sexes. This notion is further supported by previous observations that *cyp19a1b* expression in the brain is minimal during embryonic development (Okubo et al., 2011, J Neuroendocrinol, 23:412–423).

The findings of Lopez and Alward (2024) pertain to the regulation of *cyp19a1b* expression by androgen receptors. While this represents an important aspect of neuroendocrine regulation, it does not appear to be directly relevant to our discussion on *cyp19a1b*-mediated regulation of androgen receptor expression.

To ensure the reliability of behavioral analyses in mutant fish, we consider a phenotype valid only when it is consistently observed in two independent mutant lines. In the mating behavior test examining *esr2a*-deficient males using *esr2b*-deficient females as stimulus females, Δ8 line males exhibited a shorter latency to initiate following than wild-type males, whereas Δ4 line males did not. This discrepancy led us to refrain from drawing conclusions about the role of *esr2a* in mating behavior, even though the mating behavior test using wild-type females as stimulus females yielded consistent results in the Δ8 and Δ4 lines. Therefore, we do not consider the reviewer's concern to be a significant issue.

(6) Not sure what's going on with the statistics, but it is not appropriate here to treat a "control" group as special. All groups are "experimental" groups. There is nothing special about the control group in this context. all should be Bonferroni post-hoc tests.

Line 619: As detailed in Response to reviewer #1's comment 7 on weaknesses, we consider Dunnett's test the most appropriate choice for the experiments presented in Figures 4C and 4E. We acknowledge that the reviewer's concern may stem from the phrase “comparisons between control and experimental groups” in the Materials and Methods section. To clarify this point, we have revised it to “comparisons between untreated and E2-treated groups in Fig. 4, C and D” for clarity.

Minor comments:

Line 47: then how can you say the aromatization hypothesis is "correct"? it only applies to a few species so far. Need to change the framing, not state so strongly such a vague thing as a hypothesis being "correct".

Line 45: To address this concern, we have modified “widely accepted as correct” to “widely acknowledged”, ensuring a more precise characterization.

Figure 1: looks like a dosage effect in males but not females. this should be discussed at some point, even if just to mention a dosage effect exists and put it in context.

Line 91: We have revised the sentence “In males, brain E2 in heterozygotes (*cyp19a1b*^{+/-}) was also reduced to 45% of the level in wild-type siblings (P = 0.0284) (Fig. 1A)” by adding “, indicating a dosage effect of *cyp19a1b* mutation” to make this point explicit.

Were male cyp19 KO aggressive towards females?

We have not observed *cyp19a1b*-deficient males exhibiting aggressive behavior towards females in our experiments. Therefore, we do not consider them aggressive toward females.

Please explain how infertility would lead to reduced mating.

Line 142: As the reviewer has questioned, even if *cyp19a1b*-deficient males exhibit infertility due to efferent duct obstruction, it is difficult to imagine that this directly leads to reduced mating. However, the inability to release sperm could indirectly affect behavior. To address this, we have added “, possibly due to the perception of impaired sperm release” after “If this is also the case in medaka, the observed behavioral defects might be secondary to infertility.”

Describe something about the timing of the treatment here. How can peripheral E2 injections restore it when peripheral levels are normal? Did these injections restore central levels? This needs to be shown experimentally.

Line 517: As described in the Materials and Methods, E2 treatment was conducted by immersing fish in E2-containing water for 4 days. However, we had not explicitly stated that the water was changed daily to maintain the nominal concentration. To clarify this and address reviewer #2's comment 9, we have revised “males were treated with 1 ng/ml of E2 (Fujifilm Wako Pure Chemical, Osaka, Japan) or vehicle (ethanol) alone by immersion in water for 4 days” to “males were treated with 1 ng/ml of E2 (Fujifilm Wako Pure Chemical, Osaka, Japan), which was first dissolved in 100% ethanol (vehicle), or with the vehicle alone by immersion in water for 4 days, with daily water changes to maintain the nominal concentration.”

Line 522: The treatment effectively restored mating activity and ara/arb expression in the brain, suggesting a sufficient increase in brain E2 levels. However, we did not measure the actual increase, and its extent remains uncertain. To reflect this in the manuscript, we have now added the following sentence: “Although the exact increase in brain E2 levels following E2 treatment was not quantified, the observed positive effects on behavior and gene expression suggest that it was sufficient.”

I know the nomenclature differs among those who study teleosts, but it's ARa and then gene is ar1 (as an example; arb would be ar2). You're recommended the following citation to remain consistent:

Munley, K. M., Hoadley, A. P., & Alward, B. A. (2023). A phylogenetics-based nomenclature system for steroid receptors in teleost fishes. General and Comparative Endocrinology, 114436.

Paralogous genes resulting from the third round of whole-genome duplication in teleosts are typically designated by adding the suffixes “a” and “b” to their gene symbols. This convention also applies to the two androgen receptor genes, commonly referred to as ara and arb. While the alternative names ar1 and ar2 may gain broader acceptance in the future, ara and arb remain more widely used at present. Therefore, we have chosen to retain ara and arb in this manuscript.

Line 268: how is this "suggesting" less aggression? They literally showed fewer aggressive displays, so it doesn't suggest it - it literally shows it.

Line 285: Following this thoughtful suggestion, we have changed “suggesting less aggression” to “showing less aggression.”

Line 317: how can you still call it the primary driver?

The stimulatory effects of aromatase/estrogens on male-typical behaviors are exerted through the potentiation of androgen/AR signaling. Thus, we still believe that androgens—specifically 11KT in teleosts—serve as the primary drivers of these behaviors.

| *Line 318: not all deficits, like aggression, were rescued.*

Line 334: To address this comment, “These behavioral deficits were rescued by estrogen administration, indicating that reduced levels of neuroestrogens are the primary cause of the observed phenotypes: in other words, neuroestrogens are pivotal for male-typical behaviors in teleosts” has been modified and now reads “Deficits in mating were rescued by estrogen administration, indicating that reduced brain estrogen levels are the primary cause of the observed mating impairment; in other words, brain-derived estrogens are pivotal at least for male-typical mating behaviors in teleosts.”

| *Line 324: what do you mean by "sufficient"? To show that, you'd have to castrate the male and only give estrogen back. the authors continue to overstate virtually every aspect of their study, seemingly in an unnecessary manner.*

Line 341: Our intention was to convey that brain-derived estrogens early in life are not essential for the expression of male-typical behaviors in teleosts. However, we recognize that the term “sufficient” could be misinterpreted as implying that estrogens alone are adequate, without contributions from other factors such as androgens. To clarify this, we have revised the text from “neuroestrogen activity in adulthood is sufficient for the execution of male-typical behaviors, while that in early in life is not requisite. Thus, while” to “brain-derived estrogens early in life is not essential for the execution of male-typical behaviors. While.”

| *Line 329: so? in adult mice, amygdala aromatase neurons still regulate aggression. The amount in adulthood seems less important compared to site-specific functions.*

Line 346: We do not intend to suggest that brain aromatase activity in adulthood plays a negligible role in male behaviors in rodents, as we have already acknowledged its necessity in the Introduction (lines 42–43). To enhance clarity and prevent misinterpretation, we have added “, although it remains important for male behavior in adulthood” to the end of the sentence: “brain aromatase activity in rodents reaches its peak during the perinatal period and thereafter declines with age.”

| *Line 351: This contradicts what you all have been saying.*

Line 65: As mentioned in Response to reviewer #1’s comment 3 on weaknesses, the following text has been added to the Introduction: “It is worth mentioning that systemic administration of estrogens and an aromatase inhibitor increased and decreased male aggression, respectively, in several teleost species, potentially reflecting the behavioral effects of brain-derived estrogens (21–24)”, providing an overview of previous studies on the effects of estrogens and aromatase on male fish aggression (Hallgren et al., 2006; O’Connell and Hofmann, 2012; Huffman et al., 2013; Jalabert et al., 2015). With this revision, we believe the inconsistency has been addressed.

Line 367: Additionally, we have revised the sentence from “treatment of males with an aromatase inhibitor reduces their male-typical behaviors (31–33)” to “treatment of males with an aromatase inhibitor reduces their male-typical behaviors, while estrogens exert the opposite effect (21–24).”

| *Line 360: change to "...possibility that is not mutually exclusive,"*

Line 378: We have revised the phrase as suggested from “Another possibility, not mutually exclusive,” to “Another possibility that is not mutually exclusive.”

| *Line 363: but it didn't rescue aggression*

Line 381: In response, we have revised the sentence from “This possibility is supported by the present observation that estrogen treatment facilitated mating behavior in *cyp19a1b*-deficient males but not in their wild-type siblings” to “This possibility is at least likely for mating behavior, as estrogen treatment facilitated mating behavior in *cyp19a1b*-deficient males but not in their wild-type siblings.”

| *Line 367: on average*

To explain the sex differences in the role of aromatase, what about the downstream molecular or neural targets? In mammals, hodology is related to sex differences. there could be convergent sex differences in regulating the same type of behaviors as well.

Our findings demonstrate that brain-derived estrogens promote the expression of *ara*, *arb*, and their downstream target genes *vt* and *gal* in males, while enhancing the expression of *npba*, a downstream target of *Esr2b* signaling, in females. The identity of additional target genes and their roles in specific neural circuits remain to be elucidated, and we aim to address these in future research.

| *Lines 378-382: this doesn't logically follow. pgf2a could be the target of estrogens which in the intact animal do regulate female sexual receptivity. And how can you say this given that your lab has shown in *esr2b* mutants females don't mate?*

We agree that PGF2 α signaling may be activated by estrogen signaling, as stated in lines 404–407: “the present finding provides a likely explanation for this apparent contradiction, namely, that neuroestrogens, rather than or in addition to ovarian-derived circulating estrogens, may function upstream of PGF2 α signaling to mediate female receptivity.” The observation that *esr2b*-deficient females do not accept male courtship is also stated in lines 401–403: “we recently challenged it by showing that female medaka deficient for *esr2b* are completely unreceptive to males, and thus estrogens play a critical role in female receptivity.”

| *Line 396-397: or the remaining estrogens are enough to activate *esr2b*-dependent female-typical mating behaviors.*

We agree that *cyp19a1b* deficiency did not completely preclude female mating behavior, most likely because residual estrogens in the brains of *cyp19a1b*-deficient females enable weak activation of *Esr2b* signaling. However, the relevant section in the Discussion is not focused on examining why mating behavior persisted, but rather on considering the implications of this finding for the neural circuits regulating mating behavior. Therefore, incorporating the suggested explanation here would shift the focus and would not be appropriate.

| *Line 420-421: this is a lot of variation. Was age controlled for?*

The time required for medaka to reach sexual maturity varies with rearing density and food availability. Due to space constraints, we adjust these parameters as needed, which led to variation in the ages of the experimental fish. However, since all experiments were conducted using sibling fish of the same age that had just reached sexual maturity, we believe this does not affect our conclusions.

| *Line 457: have these kits been validated in medaka?*

Although we have not directly validated its applicability in medaka, its extensive use in this species suggests that it is unlikely to pose any issues (e.g., Ussery et al., 2018, *Aquat Toxicol*, 205:58–65; Lee et al., 2019, *Ecotoxicol Environ Saf*, 173:174–181; Kayo et al., 2020, *Gen Comp Endocrinol*, 285:113272; Fischer et al., 2021, *Aquat Toxicol*, 236:105873; Royan et al., 2023, *Endocrinology*, 164:bqad030).

Line 589, re fish that spawned: how many times did this happen? Please note it is based on genotype and experiment. This could be important.

Line 627: In response to this comment, we have added the following details: “Specifically, 7/18 *cyp19a1b*^{+/+}, 11/18 *cyp19a1b*^{+/-}, and 6/18 *cyp19a1b*^{-/-} males were excluded in Fig. 1D; 6/10 *cyp19a1b*^{+/+}, 3/10 *cyp19a1b*^{+/-}, and 6/10 *cyp19a1b*^{-/-} females were excluded in Fig. 6B; 2/23 *esr1*^{+/+} and 5/24 *esr1*^{-/-} males were excluded in Fig. S7; 2/24 *esr2a*^{+/+} and 3/23 *esr2a*^{-/-} males were excluded in Fig. S8A; 0/23 *esr2a*^{+/+} and 0/23 *esr2a*^{-/-} males were excluded in Fig. S8B.”

Reviewer #2 (Recommendations For The Authors):

Abstract:

(A1) The framing of neuroestrogens being important for male-typical rodents, and not for other vertebrate lineages, does not account for other groups (birds) in which this is true (the authors can consult their cited work by Balthazart (Reference 6) for extensive accounting of this). This makes the novelty clause in the abstract "indicating that neuroestrogens are pivotal for male-typical behaviors even in nonrodents" less surprising and should be acknowledged by the authors by amending or omitting this novelty clause. The findings regarding androgen receptor transcription (next sentence) are more important and pertinent.

Line 27: We recognize that the aromatization hypothesis applies to some birds, including zebra finches, as stated in the Introduction (lines 48–49) and Discussion (lines 432–433). However, this was not reflected in the Abstract. Following the reviewer’s suggestion, we have changed “in non-rodents” to “in teleosts.”

(A2) The medaka line that has been engineered to have aromatase absent in the brain is presented briefly in the abstract, but can be misinterpreted as naturally occurring. This should be amended, by including something like "engineered" or "directed mutant" before 'male medaka fish'.

Line 24: We have added “mutagenesis-derived” before “male medaka fish” in response to this comment.

Introduction:

*(I1) The paragraph on teleost brain aromatase should acknowledge that while the capacity for estrogen synthesis in the brain is 100-1000 fold higher in teleosts as compared to rodents and other vertebrates, the majority of this derives from glial and not neural sources. This can be confusing for readers since the term 'neuroestrogens' often refers to the neuronal origin and signalling. And this observation includes the exclusive radial glial expression of *cyp19a1b* in medaka (Diotel et al., 2010), and first discovered in midshipman (Forlano et al., 2001), each of which should also be cited here. In addition, the authors expend much text comparing teleosts and rodents, but it is worth expanding these kinds of comparisons, especially by pointing out that parts of the primate brain are found to densely express aromatase (see work by Ei Terasawa and others).*

In response to this comment and a similar comment from reviewer #1, we have replaced “neuroestrogens” with “brain-derived estrogens” or “brain estrogens” throughout the manuscript.

Line 63: We have also added the text “In teleost brains, including those of medaka, aromatase is exclusively localized in radial glial cells, in contrast to its neuronal localization in rodent brains (18–20).” As a result of this addition, we have changed “This observation suggests” to “These observations suggest” in the subsequent sentence.

Line 51: Additionally, to include information on aromatase in the primate brain, we have added the following text: “In primates, the hypothalamic aromatization of androgens to estrogens plays a central role in female gametogenesis (10) but is not essential for male behaviors (7, 8).”

The following references (#10 and 18–20), cited in the newly added text above, have been included in the reference list, with other references renumbered accordingly:

E. Terasawa, Neuroestradiol in regulation of GnRH release. *Horm. Behav.* 104, 138–145 (2018).

P. M. Forlano, D. L. Deitcher, D. A. Myers, A. H. Bass, Anatomical distribution and cellular basis for high levels of aromatase activity in the brain of teleost fish: aromatase enzyme and mRNA expression identify glia as source. *J. Neurosci.* 21, 8943–8955 (2001).

N. Diotel, Y. Le Page, K. Mouriec, S. K. Tong, E. Pellegrini, C. Vaillant, I. Anglade, F. Brion, F. Pakdel, B. C. Chung, O. Kah, Aromatase in the brain of teleost fish: expression, regulation and putative functions. *Front. Neuroendocrinol.* 31, 172–192 (2010).

A. Takeuchi, K. Okubo, Post-proliferative immature radial glial cells female-specifically express aromatase in the medaka optic tectum. *PLoS One* 8, e73663 (2013).

(I2) It is difficult to resolve from the introduction and work cited how restricted cyp19a1b is to the medaka brain. Important for the results of this study, it is not clear whether it is more of a bias in the brain vs other tissues, or if the cyp19a1b deficiency is restricted to the brain, and gonadal/peripheral cyp19 expression persists. The authors need to improve their consideration of the alternatives, i.e., that this manipulation is not somehow affecting: 1) peripheral aromatase expression (either cyp19a1a or cyp19a1b) in the gonad or elsewhere, 2) compensatory processes, such as other steroidogenic genes (are androgen synthesizing enzymes increasing?).

Our previous study demonstrated that *cyp19a1b* is expressed in the gonads, but at levels tens to hundreds of times lower than those in the brain (Okubo et al., 2011, *J Neuroendocrinol* 23:412–423). Additionally, a separate study in medaka reported that *cyp19a1b* expression in the ovary is considerably lower than that of *cyp19a1a* (Nakamoto et al., 2018, *Mol Cell Endocrinol* 460:104–122). Given these observations, any potential effect of *cyp19a1b* knockout on peripheral estrogen synthesis is likely negligible. Indeed, Figures S1C and S1D confirm that *cyp19a1b* knockout does not alter peripheral E2 levels.

Line 72: To incorporate this information into the Introduction and address the following comment, we have added the following text: “In medaka, *cyp19a1b* is also expressed in the gonads, but only at a level tens to hundreds of times lower than in the brain and substantially lower than that of *cyp19a1a* (26, 27).”

The following references (#26 and 27), cited in the newly added text above, have been included in the reference list, with other references renumbered accordingly:

K. Okubo, A. Takeuchi, R. Chaube, B. Paul-Prasanth, S. Kanda, Y. Oka, Y. Nagahama, Sex differences in aromatase gene expression in the medaka brain. *J. Neuroendocrinol.* 23, 412–423 (2011).

M. Nakamoto, Y. Shibata, K. Ohno, T. Usami, Y. Kamei, Y. Taniguchi, T. Todo, T. Sakamoto, G. Young, P. Swanson, K. Naruse, Y. Nagahama, Ovarian aromatase loss-of-function mutant medaka undergo ovary degeneration and partial female-to-male sex reversal after puberty. *Mol. Cell. Endocrinol.* 460, 104–122 (2018).

We have not assessed whether the expression of other steroidogenic enzymes is altered in *cyp19a1b* deficient fish, and this may be investigated in future studies.

(I3) Related, there are documented sex differences in the brain expression of cyp19a1b especially in adulthood (Okubo et al 2011) and this study should be cited here for context.

Line 72: As stated in our previous response, we have cited Okubo et al. (2011) by adding the following sentence: “In medaka, *cyp19a1b* is also expressed in the gonads, but only at a level tens to hundreds of times lower than in the brain and substantially lower than that of *cyp19a1a* (26, 27).”

Methods

(M1) The rationale is unclear as presented for using mutagen screening for cyp19a1b while using CRISPR for esr2a. Are there methodological/biochemical reasons why the authors chose to not use the same method for both?

At the time we generated the *cyp19a1b* knockouts, genome editing was not yet available, and the TILLING-based screening was the only method for obtaining mutants in medaka. In contrast, by the time we generated the *esr2a* knockouts, CRISPR/Cas9 had become available, enabling a more efficient and convenient generation of knockout lines. This is why the two knockout lines were generated using different methods.

(M2) Measurement of steroids in biological matrices is not straightforward, and it is good that the authors use multiple extraction steps (organic followed by C18 columns) before loading samples on the ELISA plates, which are notoriously sensitive. Even though these methods have been published before by this group of authors previously, the quality control and ELISA performance values (recovery, parallelism, etc.) should be presented for readers to evaluate.

Thank you for appreciating our sample purification method. Unfortunately, we have not evaluated the recovery rate or parallelism, but we recognize this a subject for future studies.

(M3) Mating behavior - E2 treated males were not co-housed with social partners for the full 24 hr before testing, but instead a few hours (?) prior to testing. The rationale for this should be spelled out explicitly.

Line 494: In response to this comment, we have added “to ensure the efficacy of E2 treatment” to the end of the sentence “The set-up was modified for E2-treated males, which were kept on E2 treatment and not introduced to the test tanks until the day of testing.”

(M4) The E2 treatment is listed as 1ng/ml vs. vehicle (ethanol). Is the E2 dissolved in 100% ethanol for administration to the tank water? Clarification is needed.

Line 517: As the reviewer correctly assumed, E2 was first dissolved in 100% ethanol before being added to the tank water. To provide this information and address reviewer #1's minor comment 5, we have revised "males were treated with 1 ng/ml of E2 (Fujifilm Wako Pure Chemical, Osaka, Japan) or vehicle (ethanol) alone by immersion in water for 4 days" to "males were treated with 1 ng/ml of E2 (Fujifilm Wako Pure Chemical, Osaka, Japan), which was first dissolved in 100% ethanol (vehicle), or with the vehicle alone by immersion in water for 4 days, with daily water changes to maintain the nominal concentration."

(M5) The authors exclude fish from the analysis of courtship display behavior for those individuals that spawned immediately at the start of the testing (and therefore it was impossible to register courtship display behaviors). How often did fish in the various treatment groups exhibit this "fast spawning" behavior? Was the occurrence rate different by treatment group? It is unlikely that these omissions from the data set drove large-scale patterns, but an indication of how often this occurred would be reassuring.

Line 627: In response to this comment, we have included the following details: "Specifically, 7/18 *cyp19a1b*^{+/+}, 11/18 *cyp19a1b*^{sup+/-}, and 6/18 *cyp19a1b*^{-/-} males were excluded in Fig. 1D; 6/10 *cyp19a1b*^{+/+}, 3/10 *cyp19a1b*^{+/-}, and 6/10 *cyp19a1b*^{-/-} females were excluded in Fig. 6B; 2/23 *esr1*^{+/+} and 5/24 *esr1*^{-/-} males were excluded in Fig. S7; 2/24 *esr2a*^{+/+} and 3/23 *esr2a*^{-/-} males were excluded in Fig. S8A; 0/23 *esr2a*^{+/+} and 0/23 *esr2a*^{-/-} males were excluded in Fig. S8B." These data indicate that the proportion of excluded males is nearly constant within each trial and is independent of the genotype of the focal fish.

Results

(R1) It is striking to see the genetic-'dose' dependent suppression of brain E2 content by heterozygous and homozygous cyp19a1b deficiency, indicating that, as the authors point out, the majority of E2 in the male medaka brain (and 1/2 in the female brain) have a brain-derived origin. It is important also for the interpretation that there are large compensatory increases in brain levels of androgens, when E2 levels drop in the cyp19a1b mutant homozygotes. This latter point should receive more attention.

Also, there are large increases in peripheral androgen levels in the homozygote mutants for cyp19a1b in both males and females. This indicates a peripheral effect in addition to the clear brain knockdown of E2 synthesis. These nuances need to be addressed.

In response to this comment, we have revised the Results section as follows:

Line 91: ", indicating a dosage effect of *cyp19a1b* mutation" has been added to the end of the sentence "In males, brain E2 in heterozygotes (*cyp19a1b*^{+/+}) was also reduced to 45% of the level in wild-type siblings (P = 0.0284) (Fig. 1A)."

Line 94: To draw more attention to the increase in brain androgen levels caused by *cyp19a1b* deficiency, "Brain levels of testosterone" has been modified to "Strikingly, brain levels of testosterone."

Line 100: "Their peripheral 11KT levels also increased 3.7- and 1.8-fold, respectively (P = 0.0789, males; P = 0.0118, females) (Fig. S1, C and D)" has been modified and now reads "In addition, peripheral 11KT levels in *cyp19a1b*^{-/-} males and females increased 3.7- and 1.8-fold, respectively (P = 0.0789, males; P = 0.0118, females) (Fig. S1, C and D), indicating peripheral influence in addition to central effects."

(R2) The interpretation on page 4 that cyp19a1b deficient males are 'less motivated' to mate is premature, given the behavioral measures used in this study. There are several competing explanations for these findings (e.g., alterations in motivation, sensory

discrimination, preference, etc.) that could be followed up in future work, but the current results are not able to distinguish among these possibilities.

Line 112: We agree that the possibility of altered cognition or sexual preference cannot be dismissed. To incorporate this perspective, we have revised the text “, suggesting that they are less motivated to mate” to “These results suggest that they are less motivated to mate, though an alternative interpretation that their cognition or sexual preference may be altered cannot be dismissed.”

(R3) On page 5, the authors present that peripheral E2 manipulation (delivery to the fish tank) restores courtship behavior in males, and then go on to erroneously conclude that this demonstrates “that reduced E2 in the brain was the primary cause of the mating defects, indicating a pivotal role of neuroestrogens in male mating behavior.” Because this is a peripheral E2 treatment, there can be manifold effects on gonadal physiology or other endocrine events that can have indirect effects on the brain and behavior. Without manipulation of E2 directly to the brain to ‘rescue’ the cyp19a1b deficiency, the authors cannot conclude that these effects are directly on the central nervous system. Tellingly, the tank E2 treatment did not rescue aggressive behavior, suggestive of the potential for indirect effects.

Line 155: As detailed in Response to reviewer #2’s specific comment 1, we have revised the text from “These results demonstrated that reduced E2 in the brain was the primary cause of the mating defects, indicating a pivotal role of neuroestrogens in male mating behavior. In contrast” to “These results suggest that reduced E2 in the brain is the primary cause of the mating defects, highlighting a pivotal role of brain-derived estrogens in male mating behavior. However, caution is warranted, as an indirect peripheral effect of bath-immersed E2 on behavior cannot be ruled out, although this is unlikely given the comparable peripheral E2 levels in *cyp19a1b*-deficient and wild-type males. In contrast to mating.”

(R4) The downregulation of androgen-dependent gene expression (vasotocin in pNVT and galanin in pPMp) in the cyp19a1b deficient males (Figure 3) could be due to exceedingly high levels of brain androgens in the cyp19a1b deficient males. The best way to test the idea that estrogens can restore the expression to be more wild-type directly (like what is happening for ara and arb) is to look at these same markers (vasotocin and galanin) in these same brain areas in the brains of E2-treated males. The authors should have these brains from Figure 2. Unless I missed something, those experiments were not performed/reported here. It is clear that the ara and arb receptors have EREs and are ‘rescued’ by E2 treatment, but in principle, there could be indirect actions for reasons stated above for the behavior due to the peripheral E2 tank application.

Thank you for your insightful comment. We agree that the current results cannot exclude the possibility that excessive androgen levels caused the downregulation of vt and gal. However, our previous studies showed that excessive 11KT administration to gonadectomized males and females increased the expression of these genes to levels comparable to wild-type males (Yamashita et al., 2020, eLife, 9:e59470; Kawabata-Sakata et al., 2024, Mol Cell Endocrinol 580:112101), making this scenario unlikely. That said, testing whether estrogen treatment restores vt and gal expression in *cyp19a1b*deficient males would be informative, and we see this as an important direction for future research.

Discussion

(D1) The authors need to clarify whether EREs are found in other vertebrate AR introns, or is this unique to the teleost genome duplication?

We have identified multiple ERE-like sequences within intron 1 of the mouse AR gene. However, sequence data alone do not provide sufficient evidence of their functionality, rendering this information of limited relevance. Therefore, we have chosen not to include this discussion in the current paper.

Reviewer #3 (Recommendations For The Authors):

(1) The authors are strongly encouraged to report information regarding the effect of Cyp19a1b deletion on the brain content of aromatase protein (ideally both isoforms investigated separately) as the two isoforms are mostly but not completely brain vs gonad specific. The analysis of other tissues would also strengthen the characterization of this model.

We agree that measuring aromatase protein levels in the brain of our fish would be valuable for confirming the loss of *cyp19a1b* function. However, as no suitable method is currently available, this issue will need to be addressed in future studies. While this constitutes indirect evidence, the observed reduction in brain E2 levels, with no change in peripheral E2 levels, in *cyp19a1b*-deficient fish strongly suggests the loss of *cyp19a1b* function, as noted in Response to reviewer #3's comment 1 on weaknesses.

(2) As presented, this study reads as niche work. A better description of the behavior and reproductive significance of the different aspects of the behavioral sequence would allow a better understanding of the results and would thus allow the non-specialist to appreciate the significance of the observations.

Line 103: In response to this comment and Reviewer #3's comment 2 on weaknesses, we have revised the sentence from "The mating behavior of medaka follows a stereotypical pattern, wherein a series of followings, courtship displays, and wrappings by the male leads to spawning" to "The mating behavior of medaka follows a stereotypical sequence. It begins with the male approaching and closely following the female (following). The male then performs a courtship display, rapidly swimming in a circular pattern in front of the female. If the female is receptive, the male grasps her with his fins (wrapping), culminating in the simultaneous release of eggs and sperm (spawning)" in order to provide a more detailed description of medaka mating behavior.

(3) The data regarding female behavior are limited and incomplete. It is suggested to keep this for another manuscript unless data on the behavior of the female herself is added. Indeed, analyzing female's behavior from the male's perspective complicates the interpretation of the results while a description of what the females do would provide valuable and interpretable information.

We thank the reviewer for this thoughtful suggestion and agree that the data and discussion for females are less extensive than for males. However, we have previously elucidated the mechanism by which estrogen/Esr2b signaling promotes female mating behavior (Nishiike et al., 2021). Accordingly, it follows that the new insights into female behavior gained from the *cyp19a1b* knockout model are more limited than those for males. Nevertheless, when combined with our prior findings, the female data in this study offer valuable insights, and the overall mechanism through which estrogens promote female mating behavior is becoming clearer. Therefore, we do not consider the female data in this study to be incomplete or merely supplementary.

(4) In Figure 2, the validity to run multiple T-tests rather than a two-way ANOVA comparing TRT and genotype is questionable. Moreover, why are the absolute values in CTL higher than in the initial experiment comparing genotypes for ara in PPa, pPPp, and

NVT as well as for arb in aPPp. More importantly, these graphs do not seem to reproduce the genotype effects for ara in pPPp and NVT and for arb in aPPp.

The data in Figures 2J and 2K were analyzed with an exclusive focus on the difference between vehicle-treated and E2-treated males, without considering genotype differences. Therefore, the use of T-tests for significance testing is appropriate.

As the reviewer noted, the overall ara expression area is larger in Figure 2J than in Figure 2F. However, as detailed in Response to reviewer #3's comment 8 on weaknesses, the relative area ratios of ara expression among brain nuclei are consistent between the two figures, indicating the reproducibility of the results. Thus, we consider this difference unlikely to affect the conclusions of this study.

Additionally, the differences in ara expression in pPPp and arb expression in aPPp between wild-type and *cyp19a1b*-deficient males appear smaller in Figures 2J and 2K compared to Figures 2F and 2H. This is likely due to the smaller sample size used in the experiments for Figures 2J and 2K, which makes the differences less distinct. However, since the same genotype-dependent trends are observed in both sets of figures, the conclusion that ara and arb expression is reduced in *cyp19a1b*-deficient male brains remains valid.

(5) More information is required regarding the analysis of single ISH - How was the positive signal selected from the background in the single ISH analyses? How was this measure standardized across animals? How many sections were imaged per region? Do the values represent unilateral or bilateral analysis?

Line 540: Following this comment, we have provided additional details on the single ISH method in the manuscript. Specifically, “, and the total area of signal in each brain nucleus was calculated using Olyvia software (Olympus)” has been revised to “The total area of signal across all relevant sections, including both hemispheres, was calculated for each brain nucleus using Olyvia software (Olympus). Images were converted to a 256-level intensity scale, and pixels with intensities from 161 to 256 were considered signals. All sections used for comparison were processed in the same batch, without corrections between samples.”

(6) More information should be provided in the methods regarding the image analysis of double ISH. In particular, what were the criteria to consider a cell as labeled are not clear. This is not clear either from the representative images.

Line 596: To provide additional details on the single ISH method in the manuscript, we have added the following sentence: “Cells were identified as coexpressing the two genes when Alexa Fluor 555 and fluorescein signals were clearly observed in the cytoplasm surrounding DAPI-stained nuclei, with intensities markedly stronger than the background noise.”

(7) There is no description of the in silico analyses run on ESR2a in the methods.

The method for identifying estrogen-responsive element-like sequences in the *esr2a* locus is described in line 549: “Each nucleotide sequence of the 5'-flanking region of *ara* and *arb* was retrieved from the Ensembl medaka genome assembly and analyzed for potential canonical ERE-like sequences using Jaspar (version 5.0_alpha) and Match (public version 1.0) with default settings.”

However, the method for domain identification in *Esr2a* was not described. Therefore, we have added the following text in line 469: “The DNA- and ligand-binding domains of medaka *Esr2a* were identified by sequence alignment with yellow perch (*Perca flavescens*) *Esr2a*, for which these domain locations have been reported (58).”

The following reference (#58), cited in the newly added text above, have been included in the reference: S. G. Lynn, W. J. Birge, B. S. Shepherd, Molecular characterization and sex-specific tissue expression of estrogen receptor α (*esr1*), estrogen receptor β (*esr2a*) and ovarian aromatase (*cyp19a1a*) in yellow perch (*Perca flavescens*). Comp. Biochem. Physiol. B Biochem. Mol. Biol. 149, 126–147 (2008).

(8) Information about the validation steps of the EIA that were carried out as well as the specificity of the antibody the steroids and the extraction efficacy should be provided.

We have not directly validated the applicability of the EIA kit, but its extensive use in medaka suggests that it is unlikely to pose any issues (e.g., Ussery et al., 2018, Aquat Toxicol, 205:58–65; Lee et al., 2019, Ecotoxicol Environ Saf, 173:174–181; Kayo et al., 2020, Gen Comp Endocrinol, 285:113272; Fischer et al., 2021, Aquat Toxicol, 236:105873; Royan et al., 2023, Endocrinology, 164:bqad030).

The specificity (cross-reactivity) of the antibodies is detailed as follows.

(1) Estradiol ELISA kits: estradiol, 100%; estrone, 1.38%; estriol, 1.0%; 5 α -dihydrotestosterone, 0.04%; androstenediol, 0.03%; testosterone, 0.03%; aldosterone, <0.01%; cortisol, <0.01%; progesterone, <0.01%.

(2) Testosterone ELISA kits: testosterone, 100%; 5 α -dihydrotestosterone, 27.4%; androstenedione, 3.7%; 11-ketotestosterone, 2.2%; androstenediol, 0.51%; progesterone, 0.14%; androsterone, 0.05%; estradiol, <0.01%.

(3) 11-Keto Testosterone ELISA kits: 11-ketotestosterone, 100%; adrenosterone, 2.9%; testosterone, <0.01%.

As this information is publicly available on the manufacturer's website, we deemed it unnecessary to include it in the manuscript.

Unfortunately, we have not evaluated the extraction efficacy of the samples, but we recognize this a subject for future studies.

(9) I wonder whether the evaluation of the impact of the mutation by comparing the behavior of a group of wild-type males to a group of mutated males is the most appropriate. Justifying this approach against testing the behavior of one mutated male facing one or several wild-type males would be appreciated.

We agree that the resident-intruder test, in which a single focal resident is confronted with one or more stimulus intruders, is the most commonly used method for assessing aggression. However, medaka form shoals and lack strong territoriality, and even slight dominance differences between the resident and the intruder can increase variability in the results, compromising data consistency. Therefore, in this study, we adopted an alternative approach: placing four unfamiliar males together in a tank and quantifying aggressive interactions in total. This method allows for the assessment of aggression regardless of territorial tendencies, making it more appropriate for our investigation.

(10) Lines 329-331: this sentence should be rephrased as it contributes to the confusion between sexual differentiation and activation of circuits. The restoration of sexual behavior by adult estrogen treatment pleads in favor of an activational role of neuro-estrogens on behavior rather than an organizational role. Therefore, referring to sexual differentiation is misleading, even more so that the study never compares sexes.

As detailed in Response to reviewer #3's comment 9 on weaknesses, we consider that all factors that cause sex differences, including the transient effects of adult steroids, need to be incorporated into a theory of sexual differentiation. In teleosts, since steroids during early development have little effect and sexual differentiation primarily relies on steroid action in adulthood, our discussion on brain sexual differentiation remains valid, including the statement in line 347: "This variation among species may represent the activation of neuroestrogen synthesis at life stages critical for sexual differentiation of behavior that are unique to each species."

(11) Lines 384-386: *I may have missed something but I do not see data supporting the notion that neuroestrogens may function upstream of PGF2α signaling to mediate female receptivity.*

Line 403: We acknowledge that our explanation was insufficient and apologize for any confusion. To clarify this point, "Given that estrogen/Esr2b signaling feminizes the neural substrates that mediate mating behavior, while PGF2α signaling triggers female sexual receptivity," has been added before the sentence "The present finding provides a likely explanation for this apparent contradiction, namely, that neuroestrogens, rather than or in addition to ovarian-derived circulating estrogens, may function upstream of PGF2α signaling to mediate female receptivity."

Additional alteration

Reference list (line 682): a preprint article has now been published in a peer-reviewed journal, and the information has been updated accordingly as follows: "bioRxiv doi: 10.1101/2024.01.10.574747 (2024)" to "Proc. Natl. Acad. Sci. U.S.A. 121, e2316459121 (2024)."

<https://doi.org/10.7554/eLife.97106.2.sa0>